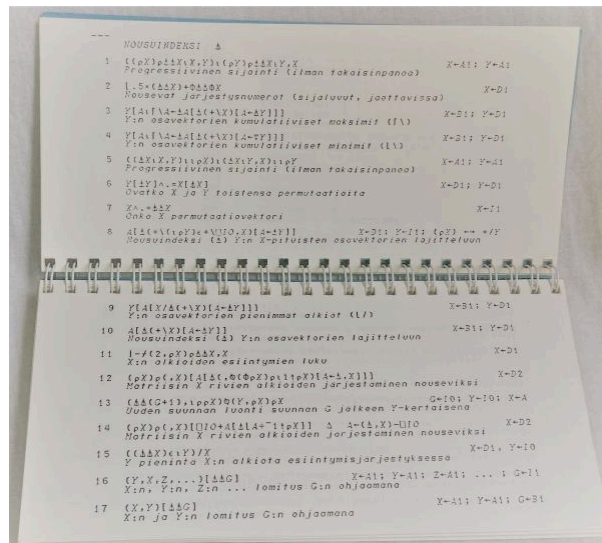


FinnAPL idiom library

The **FinnAPL idiom library** contains a collection of over 700 one-line APL idioms to accomplish a large variety of tasks. It was first presented at the 1984 APL conference in Helsinki, Finland. The huge contribution of the Finnish APL Association is gratefully acknowledged.

This listing mainly suffers from two issues:

- Due to its age, it doesn't make use of modern APL features which can provide a simpler solution. (However, a simple-looking expression which uses nested arrays might be far more computationally expensive than a longer, more involved solution which only uses simple arrays.)
- It can be hard to find what one is looking for, both because computing terminology has changed, and because pinpointing the exact term can be difficult.



The original FinnAPL idiom library

APLcart includes all of the below library, updated to use the latest language features, and in an easily searchable format that includes modern day synonyms.

Contents

Interpreting an entry in the Idiom Library

Idiom Library Listing

[Grade Up \$\Delta\$](#)

[Grade Down \$\nabla\$](#)

[Matrix Inversion / Matrix Division \$\boxdiv\$](#)

[Decode \$\perp\$](#)

[Encode \$\top\$](#)

[Logarithm \$\odot\$](#)

[Branch \$\rightarrow\$](#)

[Execute \$\downarrow\$](#)

[Format \$\overline{}\$](#)

[Roll / Deal ?](#)

[Geometrical Functions \$\circ\$](#)

[Factorial / Binomial !](#)

[Index Of \$\iota\$](#)

[Outer Product \$\circ.!\$ \$\circ.\top\$ \$\circ.!\$](#)

[Outer Product \$\circ.*\$ \$\circ.\times\$ \$\circ.-\$ \$\circ.+\$](#)

[Outer Product \$\circ.<\$ \$\circ.\leq\$ \$\circ.\geq\$ \$\circ.>\$](#)

[Outer Product \$\circ.\neq\$ \$\circ.=\$](#)

[Inner Product \$\lceil.\times\$ \$\lfloor.\times\$ \$\lceil.+\$ \$\times.\circ\$ \$\times.*\$ \$+.*\$](#)

Inner Product $\vee \wedge < < < \leq < \geq \leq \geq > >$ Inner Product $\vee \neq \wedge = + \neq + =$ Inner Product $- \div + \div + \times$ Scan $\backslash \backslash \backslash \times \backslash - \backslash$ Scan $\wedge \backslash < \backslash \leq \backslash \neq \backslash$ Scan $\vee \backslash \wedge \backslash$ Scan $+$ Reduction $\circ / \div / - / \times /$ Reduction $/ / \lfloor /$ Reduction $\vee / \wedge / \neq /$ Reduction $\wedge /$ Reduction $+$ Reverse $\textcircled{1} \ominus$ Rotate $\textcircled{1} \ominus$ Transpose $\textcircled{Q}$ Maximum $\lceil$ Minimum $\lfloor$ Ceiling $\lceil$ Floor $\lfloor$ Residue $\lceil$ Magnitude $\lceil$, Signum $\times$ Expand $\backslash \rightsquigarrow$ Compress $/ \neq$ Take $\uparrow$ Drop $\downarrow$ Member Of $\in$ Index Generator ι Logical Functions $\sim \vee \wedge \forall \wedge$ Comparison $< \leq > \neq$ Ravel $,$ Catenate $,$ Indexing $[]$ Shape ρ Reshape ρ Arithmetic $+$ $-$ $\times$ $\div$ Miscellaneous**Notes**

Interpreting an entry in the Idiom Library

As an example of how each entry in the library is arranged, consider the first idiom:

1.	Progressive index of (without replacement)	$X \leftarrow A1; Y \leftarrow A1$
	$((\rho X) \rho \Delta \Delta X \iota X, Y) \iota (\rho Y) \rho \Delta \Delta X \iota Y, X$	

The entry includes a brief description of what the idiom does, which is followed by the expression $X \leftarrow A1; Y \leftarrow A1$ which specifies the types and ranks of the arguments:

A	Any [Numeric, Character or Boolean]
D	Numeric
I	Integer
C	Character
B	Boolean

A number following the type indicates the rank, e.g.

A0	Any scalar (rank 0)
A1	Any vector (rank 1)
A2	Any matrix (rank 2)

Thus the idiom shown expects two character or numeric vectors, X and Y . It will find the index position of each element of Y in X , for example:

```
X←'which side does an ostrich have its feathers?'
Y←'on the outside, of course!'
((pX)ρ⍳⍳X⍳Y)⍳(pY)ρ⍳⍳X⍳Y,X
13 18 6 22 2 10 11 20 46 34 7 3 9 14 46 16 46 37 19 4 46 46 23 15 31 46
(X,'-')[((pX)ρ⍳⍳X⍳Y)⍳(pY)ρ⍳⍳X⍳Y,X]
on the o-tside- -f c--rse-
```

In this example, the first 'o' character in Y occurs in at index position 13 in X , the second one occurs at position 20, and the third and fourth 'o' characters are not present in X .

For a more detailed description of how this particular idiom works, see [this analysis \(http://www.sudleyplace.com/APL/AnatomyOfAnIdiom.ahtml\)](http://www.sudleyplace.com/APL/AnatomyOfAnIdiom.ahtml) by Bob Smith.

Idiom Library Listing

Grade Up ∇

1.	Progressive index of (without replacement)	$X \leftarrow A1; Y \leftarrow A1$
	$((\rho X) \rho \nabla X \uparrow X, Y) \uparrow (\rho Y) \rho \nabla X \uparrow Y, X$	
2.	Ascending cardinal numbers (ranking, shareable)	$X \leftarrow D1$
	$ \cdot 5 \times (\nabla X) + \phi \nabla \phi X$	
3.	Cumulative maxima ($\uparrow \backslash$) of subvectors of Y indicated by X	$X \leftarrow B1; Y \leftarrow D1$
	$Y[A \uparrow \backslash A \leftarrow \nabla A[\nabla(+\backslash X)[A \leftarrow \nabla Y]]]$	
4.	Cumulative minima ($\downarrow \backslash$) of subvectors of Y indicated by X	$X \leftarrow B1; Y \leftarrow D1$
	$Y[A \downarrow \backslash A \leftarrow \nabla A[\nabla(+\backslash X)[A \leftarrow \nabla Y]]]$	
5.	Progressive index of (without replacement)	$X \leftarrow A1; Y \leftarrow A1$
	$((\nabla X \uparrow X, Y) \uparrow \rho X) \uparrow (\nabla X \uparrow Y, X) \uparrow \rho Y$	
6.	Test if X and Y are permutations of each other	$X \leftarrow D1; Y \leftarrow D1$
	$Y[\nabla Y] \wedge . = X[\nabla X]$	
7.	Test if X is a permutation vector	$X \leftarrow I1$
	$X \wedge . = \nabla X$	
8.	Grade up (∇) for sorting subvectors of Y having lengths X	$Y \leftarrow D1; X \leftarrow I1; (\rho Y) \leftrightarrow +/X$
	$A[\nabla(+\backslash(\rho Y) \in +\backslash I0, X)[A \leftarrow \nabla Y]]$	
9.	Index of the elements of X in Y	$X \leftarrow D1; Y \leftarrow D1$
	$(((1, A) / B) [1 + \rho Y] [(\rho Y) \downarrow (+\backslash 1, A \leftarrow (1 \downarrow A) \neq 1 \downarrow A \leftarrow A[B])][\nabla B \leftarrow \nabla A \leftarrow Y, X])$	
10.	Minima ($\downarrow /$) of elements of subvectors of Y indicated by X	$X \leftarrow B1; Y \leftarrow D1$
	$Y[A[X/\nabla(+\backslash X)[A \leftarrow \nabla Y]]]$	
11.	Grade up (∇) for sorting subvectors of Y indicated by X	$X \leftarrow B1; Y \leftarrow D1$
	$A[\nabla(+\backslash X)[A \leftarrow \nabla Y]]$	
12.	Occurences of the elements of X	$X \leftarrow D1$
	$ \cdot \neq (2, \rho X) \rho \nabla X, X$	
13.	Sorting rows of matrix X into ascending order	$X \leftarrow D2$
	$(\rho X) \rho (, X)[A[\nabla(, \nabla(\phi \rho X) \rho 1 \uparrow \rho X)[A \leftarrow \nabla, X]]]$	
14.	Adding a new dimension after dimension G Y-fold	$G \leftarrow I0; Y \leftarrow I0; X \leftarrow A$
	$(\nabla(G+1), \uparrow \rho \rho X) \nabla(Y, \rho X) \rho X$	
15.	Sorting rows of matrix X into ascending order	$X \leftarrow D2$
	$A \leftarrow (\nabla, X) - \square I0 \cdot (\rho X) \rho (, X)[\square I0 + A[\nabla[A \div 1 \uparrow \rho X]]]$	
16.	Y smallest elements of X in order of occurrence	$X \leftarrow D1, Y \leftarrow I0$

	$((\Delta\Delta X) \in \iota Y) / X$	
17.	Merging X, Y, Z ... under control of G (mesh) $(X, Y, Z, \dots) [\Delta\Delta G]$	$X \leftarrow A1; Y \leftarrow A1; Z \leftarrow A1; \dots; G \leftarrow I1$
18.	Merging X and Y under control of G (mesh) $(X, Y) [\Delta\Delta G]$	$X \leftarrow A1; Y \leftarrow A1; G \leftarrow B1$
19.	Ascending cardinal numbers (ranking, all different) $\Delta\Delta X$	$X \leftarrow D1$
20.	Grade down (Ψ) for sorting subvectors of Y having lengths X $A[\Delta(+\backslash(\iota\rho Y) \in +\backslash\Box I0, X) [A \leftarrow \Psi Y]]$	$Y \leftarrow D1; X \leftarrow I1; (\rho Y) \leftrightarrow +/X$
21.	Maxima ($/$) of elements of subvectors of Y indicated by X $Y[A[X/\Delta(+\backslash X) [A \leftarrow \Psi Y]]]$	$X \leftarrow B1; Y \leftarrow D1$
22.	Grade down (Ψ) for sorting subvectors of Y indicated by X $A[\Delta(+\backslash X) [A \leftarrow \Psi Y]]$	$X \leftarrow B1; Y \leftarrow D1$
23.	Y largest elements of X in order of occurrence $((\Delta\Psi X) \in \iota Y) / X$	$X \leftarrow D1; Y \leftarrow I0$
24.	Merging X and Y under control of G (mesh) $(Y, X) [\Delta\Psi G]$	$X \leftarrow A1; Y \leftarrow A1; G \leftarrow B1$
25.	Descending cardinal numbers (ranking, all different) $\Delta\Psi X$	$X \leftarrow D1$
26.	Sorting rows of X according to key Y (alphabetizing) $X[\Delta(1+\rho Y) \perp Y \iota \Phi X;]$	$X \leftarrow A2; Y \leftarrow A1$
27.	Diagonal ravel $(, X) [\Delta+ \neq (\rho X) \top (\iota\rho, X) - \Box I0]$	$X \leftarrow A$
28.	Grade up according to key Y $\Delta Y \iota X$	$Y \leftarrow A1; X \leftarrow A1$
29.	Test if X is a permutation vector $X[\Delta X] \wedge . = \iota\rho X$	$X \leftarrow I1$
30.	Sorting a matrix into lexicographic order $X[\Delta+ \neq A < . - \Phi A \leftarrow X, 0;]$	$X \leftarrow D2$
31.	Sorting words in list X according to word length $X[\Delta X + . \neq ' ' ;]$	$X \leftarrow C2$
32.	Classification of X to classes starting with Y $A[(B/C) - \rho Y] \leftarrow B + \backslash \sim B \leftarrow (\rho Y) < C \leftarrow \Delta Y, X + A \leftarrow 0 \times X \cdot A$	$X \leftarrow D1; Y \leftarrow D1; Y < . \geq 1 \Phi Y$
33.	Rotate first elements (1ϕ) of subvectors of Y indicated by X $Y[\Delta X ++ \backslash X]$	$X \leftarrow B1; Y \leftarrow A1$
34.	Doubling quotes (for execution)	$X \leftarrow C1$

	$(X, ' '')[(\neg I0+pX) \uparrow (\neg \neg X), (' '=X)/\neg pX]$	
35.	Inserting Y * 's into vector X after indices G	$X \leftarrow C1; Y \leftarrow I0; G \leftarrow I1$
	$(X, '*')[(\neg I0+pX) \uparrow (\neg \neg X), (Y \times pG) pG]$	
36.	Median ^[1]	$X \leftarrow D1$
	$X[(\neg X)[[.5 \times pX]]]$	
37.	Index of last maximum element of X	$X \leftarrow D1$
	$\neg 1 \uparrow \neg X$	
38.	Index of (first) minimum element of X	$X \leftarrow D1$
	$1 \uparrow \neg X$	
39.	Expansion vector with zero after indices Y	$X \leftarrow D1; Y \leftarrow I1$
	$(pX) \geq \neg (\neg pX), Y$	
40.	Catenating G elements H before indices Y in vector X	$X \leftarrow A1; Y \leftarrow I1; G \leftarrow I0; H \leftarrow A0$
	$A \leftarrow G \times p, Y \cdot ((ApH), X) \uparrow (\neg (ApY), \neg pX)$	
41.	Catenating G elements H after indices Y in vector X	$X \leftarrow A1; Y \leftarrow I1; G \leftarrow I0; H \leftarrow A0$
	$A \leftarrow G \times p, Y \cdot (X, ApH) \uparrow (\neg (\neg pX), ApY)$	
42.	Merging X and Y under control of G (mesh)	$X \leftarrow A1; Y \leftarrow A1; G \leftarrow B1$
	$A[\neg G] \leftarrow A \leftarrow Y, X \cdot A$	
43.	Sorting a matrix according to Y:th column	$X \leftarrow D2$
	$X[\neg X[;Y];]$	
44.	Sorting indices X according to data Y	$X \leftarrow I1; Y \leftarrow D1$
	$X[\neg Y[X]]$	
45.	Choosing sorting direction during execution	$X \leftarrow D1; Y \leftarrow I0$
	$\neg X \times (\neg 1 \ 1)[Y]$	
46.	Sorting Y according to X	$X \leftarrow A1; Y \leftarrow A1$
	$Y[\neg X]$	
47.	Sorting X into ascending order	$X \leftarrow D1$
	$X[\neg X]$	
48.	Inverting a permutation	$X \leftarrow I1$
	$\neg X$	

Grade Down ▼

49.	Reverse vector X on condition Y	$X \leftarrow A1; Y \leftarrow B0$
	$X[\neg Y! \neg pX]$	
50.	Sorting a matrix into reverse lexicographic order	$X \leftarrow D2$
	$X[\neg \neg + \neg A < . - \neg A \leftarrow X, 0;]$	
52.	Reversal (ϕ) of subvectors of X having lengths Y	$X \leftarrow D1; Y \leftarrow I1$

	$X[\phi\psi+\backslash(\iota\rho X)\epsilon+\backslash\Box I0,Y]$	
53.	Reversal (ϕ) of subvectors of Y indicated by X	$X\leftarrow B1; Y\leftarrow A1$
	$Y[\phi\psi+\backslash X]$	
55.	Indices of ones in logical vector X	$X\leftarrow B1$
	$(+/X)\uparrow\psi X$	
56.	Index of first maximum element of X	$X\leftarrow D1$
	$1\uparrow\psi X$	
57.	Moving all blanks to end of text	$X\leftarrow C1$
	$X[\psi' \neq X]$	
58.	Sorting X into descending order	$X\leftarrow D1$
	$X[\psi X]$	
59.	Moving elements satisfying condition Y to the start of X	$X\leftarrow A1; Y\leftarrow B1$
	$X[\psi Y]$	

Matrix Inversion / Matrix Division

60.	Interpolated value of series (X,Y) at G	$X\leftarrow D1; Y\leftarrow D1; G\leftarrow D0$
	$G\downarrow Y\Box X\circ.*\phi-\Box I0-\iota\rho X$	
61.	Predicted values of exponential (curve) fit	$X\leftarrow D1; Y\leftarrow D1$
	$*A+.\times(\odot Y)\Box A\leftarrow X\circ.*0\ 1$	
62.	Coefficients of exponential (curve) fit of points (X,Y)	$X\leftarrow D1; Y\leftarrow D1$
	$A\leftarrow(\odot Y)\Box X\circ.*0\ 1\cdot A[1]\leftarrow*A[1]\cdot A$	
63.	Predicted values of best linear fit (least squares)	$X\leftarrow D1; Y\leftarrow D1$
	$A+.\times Y\Box A\leftarrow X\circ.*0\ 1$	
64.	G-degree polynomial (curve) fit of points (X,Y)	$X\leftarrow D1; Y\leftarrow D1$
	$\phi Y\Box X\circ.*0,\iota G$	
65.	Best linear fit of points (X,Y) (least squares)	$X\leftarrow D1; Y\leftarrow D1$
	$Y\Box X\circ.*0\ 1$	

Decode

66.	Binary format of decimal number X	$X\leftarrow I0$
	$\mp 10\downarrow((1+[2\odot[/\backslash,X])\rho 2)\tau X$	
67.	Barchart of two integer series (across the page)	$X\leftarrow I2; 1\rho\rho X\leftrightarrow 2$
	$'*\odot\odot'[\Box I0+2\downarrow X\circ.\geq 1[/\backslash,X]$	
68.	Case structure with an encoded branch destination	$Y\leftarrow I1; X\leftarrow B1$
	$\rightarrow Y[1+2\downarrow X]$	

69.	Representation of current time (24 hour clock)	
	$A \leftarrow \overline{\tau}1000 \downarrow 3 \uparrow 3 \downarrow \square TS \cdot A[3 \ 6] \leftarrow ' : ' \cdot A$	
70.	Representation of current date (descending format)	
	$A \leftarrow \overline{\tau}1000 \downarrow 3 \uparrow \square TS \cdot A[5 \ 8] \leftarrow ' - ' \cdot A$	
71.	Representation of current time (12 hour clock)	
	$(1\phi, ' :: ', 3 \ 2p6 \ 0\overline{\tau}100 \downarrow 12 \ 0 \ 0 3 \uparrow 3 \downarrow \square TS), 'AP'[1+12\leq \square TS[4]], 'M'$	
73.	Removing duplicate rows	$X \leftarrow A2$
	$((A \uparrow A) = \uparrow p A \leftarrow 2 \downarrow X \wedge . = \phi X) \neq X$	
74.	Conversion from hexadecimal to decimal	$X \leftarrow C$
	$16 \downarrow - \square I0 - '0123456789ABCDEF' \uparrow \phi X$	
75.	Conversion of alphanumeric string into numeric	$X \leftarrow C1$
	$10 \downarrow ^{-1} + '0123456789' \uparrow X$	
76.	Value of polynomial with coefficients Y at points X	$X \leftarrow D1; Y \leftarrow D1$
	$(X \circ . +, 0) \downarrow Y$	
77.	Changing connectivity list X to a connectivity matrix	$X \leftarrow C2$
	$A \leftarrow (\times / B \leftarrow 0 + [/ , X) p 0 \cdot A[\square I0 + B[1] \downarrow - \square I0 - X] \leftarrow 1 \cdot B p A$	
78.	Present value of cash flows X at interest rate Y %	$X \leftarrow D1; Y \leftarrow D0$
	$(\div 1 + Y \div 100) \downarrow \phi X$	
79.	Justifying right	$X \leftarrow C$
	$(1 - (' ' = X) \downarrow 1) \phi X$	
80.	Number of days in month X of years Y (for all leap years)	$X \leftarrow I0; Y \leftarrow I$
	$(12p7p31 \ 30)[X] - 0 \uparrow ^{-1} + 2 \downarrow (X=2), [.1](0 \neq 400 Y) - (0 \neq 100 Y) - 0 \neq 4 Y$	
81.	Number of days in month X of years Y (for most leap years)	$X \leftarrow I0; Y \leftarrow I$
	$(12p7p31 \ 30)[X] - 0 \uparrow ^{-1} + 2 \downarrow (X=2), [.1]0 \neq 4 Y$	
82.	Encoding current date	
	$100 \downarrow 100 3 \uparrow \square TS$	
83.	Removing trailing blanks	$X \leftarrow C1$
	$(1 - (' ' = X) \downarrow 1) \downarrow X$	
84.	Index of first non-blank, counted from the rear	$X \leftarrow C1$
	$(' ' = X) \downarrow 1$	
85.	Indexing scattered elements	$X \leftarrow A; Y \leftarrow I2$
	$(, X) [\square I0 + (pX) \downarrow Y - \square I0]$	
86.	Conversion of indices Y of array X to indices of raveled X	$X \leftarrow A; Y \leftarrow I2$
	$\square I0 + (pX) \downarrow Y - \square I0$	
87.	Number of columns in array X as a scalar	$X \leftarrow A$
	$0 \downarrow pX$	

88.	Future value of cash flows X at interest rate Y %	$X \leftarrow D1; Y \leftarrow D0$
	$(1+Y \div 100) \uparrow X$	
89.	Sum of the elements of vector X	$X \leftarrow D1$
	$1 \uparrow X$	
90.	Last element of numeric vector X as a scalar	$X \leftarrow D1$
	$0 \uparrow X$	
91.	Last row of matrix X as a vector	$X \leftarrow A$
	$0 \uparrow X$	
92.	Integer representation of logical vectors	$X \leftarrow B$
	$2 \uparrow X$	
93.	Value of polynomial with coefficients Y at point X	$X \leftarrow D0; Y \leftarrow D$
	$X \uparrow Y$	

Encode T

94.	Conversion from decimal to hexadecimal ($X=1..255$)	$X \leftarrow I$
	$\mathbb{Q}'0123456789ABCDEF' [\square I0 + (([\div 16 \ominus, X) \rho 16) \tau X]$	
	this alternative opens the range to 0.. $\div 10$	
	$\mathbb{Q}'0123456789ABCDEF' [\square I0 + ((1 + [16 \ominus [/ X + X = 0) \rho 16) \tau X]$	
95.	All binary representations up to X (truth table)	$X \leftarrow I0$
	$(([2 \ominus 1 + X) \rho 2) \tau 0, \imath X$	
96.	Representation of X in base Y	$X \leftarrow D0; Y \leftarrow D0$
	$((1 + [Y \ominus X) \rho Y) \tau X$	
97.	Digits of X separately	$X \leftarrow I0$
	$((1 + [10 \ominus X) \rho 10) \tau X$	
98.	Helps locating column positions 1..X	$X \leftarrow I0$
	$1 \ 0 \mp 10 \ 10 \tau 1 - \square I0 - \imath X$	
99.	Conversion of characters to hexadecimal representation ($\square AV$)	$X \leftarrow C1$
	$, ' ', \mathbb{Q}'0123456789ABCDEF' [\square I0 + 16 \ 16 \tau - \square I0 - \square AV \imath X]$	
100.	Polynomial with roots X	$X \leftarrow D1$
	$\phi((\theta, \imath \rho X) \circ . = + \mp \sim A) + . \times (-X) \times . * A \leftarrow ((\rho X) \rho 2) \tau^{-1} 1 + \imath 2 * \rho X$	
101.	Index pairs of saddle points	$X \leftarrow D2$
	$\square I0 + (\rho X) \tau - \square I0 - (, (X = (\rho X) \rho [\mp X) \wedge X = \mathbb{Q}(\phi \rho X) \rho [/ X) / \imath \times / \rho X$	
102.	Changing connectivity matrix X to a connectivity list	$X \leftarrow C2$
	$(, X) / 1 + A \tau^{-1} 1 + \imath \times / A \leftarrow \rho X$	
103.	Matrix of all indices of X	$X \leftarrow A$
	$\square I0 + (\rho X) \tau (\imath \times / \rho X) - \square I0$	

104.	Separating a date YYMMDD to YY, MM, DD	$X \leftarrow D$
	$\Phi(3\rho 100)T X$	
105.	Indices of elements Y in array X	$X \leftarrow A; Y \leftarrow A$
	$\lceil I0 + (\rho X)T(-\lceil I0) + (, X \in Y) / \lceil \rho, X$	
106.	All pairs of elements of $\lceil X$ and $\lceil Y$	$X \leftarrow I0; Y \leftarrow I0$
	$\lceil I0 + (X, Y)T(\lceil X \times Y) - \lceil I0$	
107.	Matrix for choosing all subsets of X (truth table)	$X \leftarrow A1$
	$((\rho X)\rho 2)T^{-1} + \lceil 2 * \rho X$	
108.	All binary representations with X bits (truth table)	$X \leftarrow I0$
	$(X\rho 2)T^{-1} + \lceil 2 * X$	
109.	Incrementing cyclic counter X with upper limit Y	$X \leftarrow D; Y \leftarrow D0$
	$1 + YTX$	
110.	Decoding numeric code ABBCCC into a matrix	$X \leftarrow I$
	$10 \ 100 \ 1000T X$	
111.	Integer and fractional parts of positive numbers	$X \leftarrow D$
	$0 \ 1TX$	

Logarithm $\otimes$

112.	Number of decimals of elements of X	$X \leftarrow D1$
	$\lceil 10 \otimes (\lceil (' . ' \neq A) / A \leftarrow \overline{T} X) \div X$	
113.	Number of sortable columns at a time using $\perp$ and alphabet X	$X \leftarrow C1$
	$\lceil (1 + \rho X) \otimes 2 * (A = \overline{1} + A \leftarrow 2 * \lceil 128) \lceil 1$	
114.	Playing order in a cup for X ranked players	$X \leftarrow I0$
	$, \Phi(A\rho 2) \rho (2 * A \leftarrow \lceil 2 \otimes X) \uparrow \lceil X$	
115.	Arithmetic precision of the system (in decimals)	
	$\lceil \lceil 10 \otimes \rceil 1 - 3 \times \div 3$	
116.	Number of digitpositions in integers in X	$X \leftarrow I$
	$1 + (X < 0) + \lceil 10 \otimes \rceil X + 0 = X$	
117.	Number of digit positions in integers in X	$X \leftarrow I$
	$1 + \lceil 10 \otimes (X = 0) + X \times (1 \overline{-} 10) \lceil 1 + X < 0 \rceil$	
118.	Number of digits in positive integers in X	$X \leftarrow I$
	$1 + \lceil 10 \otimes X + 0 = X$	

Branch $\rightarrow$

119.	Case structure according to key vector G	$X \leftarrow A0; Y \leftarrow I1; G \leftarrow A1$
------	--	---

	$\rightarrow Y[G \downarrow X]$	
120.	Forming a transitive closure	$X \leftarrow B2$
	$\rightarrow \square LC[\downarrow v / , (X \leftarrow X \vee X \vee . \wedge X) \neq +X]$	
121.	Case structure with integer switch	$X \leftarrow I0; Y \leftarrow I1$
	$\rightarrow X \phi Y$	
122.	For-loop ending construct	$X \leftarrow I0; Y \leftarrow I0; G \leftarrow I0$
	$\rightarrow Y[\downarrow G \geq X \leftarrow X+1]$	
123.	Conditional branch to line Y	$X \leftarrow B0; Y \leftarrow I0; Y > 0$
	$\rightarrow Y[\downarrow X]$	
124.	Conditional branch out of program	$X \leftarrow B0$
	$\rightarrow 0[\downarrow X]$	
125.	Conditional branch depending on sign of X	$X \leftarrow I0; Y \leftarrow I1$
	$\rightarrow Y[2 + \times X]$	
126.	Continuing from line Y (if $X > 0$) or exit	$X \leftarrow D0; Y \leftarrow I0$
	$\rightarrow Y \times \times X$	
127.	Case structure using levels with limits G	$X \leftarrow D0; G \leftarrow D1; Y \leftarrow I1$
	$\rightarrow (X \geq G) / Y$	
128.	Case structure with logical switch (preferring from start)	$X \leftarrow B1; Y \leftarrow I1$
	$\rightarrow X / Y$	
129.	Conditional branch out of program	$X \leftarrow B0$
	$\rightarrow 0 \times \downarrow X$	

Execute $\downarrow$

132.	Test for symmetricity of matrix X	$X \leftarrow A2$
	$\downarrow \downarrow '1', ' \uparrow \downarrow ' [\square I0 + \wedge / (\rho X) = \phi \rho X], ' ' ' 0 \sim 0 \in X = \phi X ' ' '$	
133.	Using a variable named according to X	$X \leftarrow A0; Y \leftarrow A$
	$\downarrow 'VAR', (\nabla X), ' \leftarrow Y'$	
134.	Rounding to $\square PP$ precision	$X \leftarrow D1$
	$\downarrow \nabla X$	
135.	Convert character or numeric data into numeric	$X \leftarrow A1$
	$\downarrow \nabla X$	
136.	Reshaping only one-element numeric vector X into a scalar	$X \leftarrow D1$
	$\downarrow \nabla X$	
137.	Graph of F(X) at points X ('X' $\in F$)	$F \leftarrow A1; X \leftarrow D1$
	$' * ' [\square I0 + (\phi(\sim 1 + \downarrow / A) + \downarrow 1 + (\downarrow / A) - \downarrow / A) \circ . = A \leftarrow \downarrow .5 + \downarrow F]$	
138.	Conversion of each row to a number (default zero)	$X \leftarrow C2$

	$(Xv.\neq ' ')\backslash 1\downarrow \downarrow '0 ' , , X, ' '$	
139.	Test for symmetricity of matrix X	$X\leftarrow A2$
	$\downarrow (\sim 7 * A \wedge . = \phi A \leftarrow \rho X) \uparrow ' 0 \sim 0 \in X = \phi X '$	
140.	Execution of expression X with default value Y	$X\leftarrow D1$
	$\downarrow ((X \wedge . = ' ') / ' Y ') , X$	
141.	Changing X if a new input value is given	$X\leftarrow A$
	$X\leftarrow \downarrow , ((2 \uparrow ' X ') , ' ' , [. 5] A) [\square I 0 + \sim ' ' \wedge . = A \leftarrow \square ;]$	
142.	Definite integral of F(X) in range Y with G steps ('X'∈F)	$F\leftarrow A1 ; G\leftarrow D0 ; Y\leftarrow D1 ; \rho Y \leftrightarrow 2$
	$A + . \times \downarrow F , 0 \rho X \leftarrow Y [1] + (A \leftarrow - - / Y \div G) \times 0 , 1 G$	
143.	Test if numeric and conversion to numeric form	$X\leftarrow C1$
	$1 \downarrow \downarrow ' 0 ' , (\wedge / X \in ' 0 1 2 3 4 5 6 7 8 9 ') / X$	
144.	Tests the social security number (Finnish)	$Y\leftarrow ' 0 1 \dots 9 A B C \dots Z ' ; 10 = \rho X$
	$(\sim 1 \uparrow X) = ((\sim Y \in ' G I O Q ') / Y) [1 + 3 1 \downarrow 9 \uparrow X]$	
145.	Conditional execution	$X\leftarrow B0$
	$\downarrow X / ' E X P R E S S I O N '$	
146.	Conditional branch out of programs	$X\leftarrow B0$
	$\downarrow X / ' \rightarrow '$	
147.	Using default value 100 if X does not exist	$X\leftarrow A$
	$\downarrow (\sim 3 * 2 \neq \square N C ' X ') \uparrow ' X 100 '$	
148.	Conditional execution	$X\leftarrow B0$
	$\downarrow X \downarrow ' \varnothing \dots '$	
149.	Giving a numeric default value for input	$X\leftarrow D0$
	$1 \rho (\downarrow \square , ' , 10 ') , X$	
150.	Assign values of expressions in X to variables named in Y	$X\leftarrow C2 ; Y\leftarrow C2$
	$A \leftarrow \downarrow , ' , ' , ' (' , ' 0 ' , ' \rho ' , Y , ' \leftarrow ' , X , ') '$	
151.	Evaluation of several expressions; results form a vector	$X\leftarrow A$
	$\downarrow , ' , ' , ' (' , ' , ' , X , ') '$	
152.	Sum of numbers in character matrix X	$X\leftarrow A2$
	$\downarrow , ' + ' , X$	
153.	Indexing when rank is not known beforehand	$X\leftarrow A ; Y\leftarrow I$
	$\downarrow ' X [' , ((\sim 1 + \rho \rho X) \rho ' ; ') , ' Y] '$	

Format ¶

154.	Numeric headers (elements of X) for rows of table Y	$X\leftarrow D1 ; Y\leftarrow A2$
	$(3 \phi 7 \ 0 \overline{} X \circ . + , 0) , \overline{} Y$	
155.	Formatting a numerical vector to run down the page	$X\leftarrow D1$

	$\overline{\tau}X^{\circ} . + , \emptyset$	
156.	Representation of current date (ascending format)	
	$A \leftarrow \overline{\tau} \phi 3 \uparrow \square TS \cdot A[(' '=A)/\iota \rho A] \leftarrow ' . ' \cdot A$	
157.	Representation of current date (American)	
	$A \leftarrow \overline{\tau} 100 1 \phi 3 \uparrow \square TS \cdot A[(' '=A)/\iota \rho A] \leftarrow ' / ' \cdot A$	
158.	Formatting with zero values replaced with blanks	$X \leftarrow A$
	$(\rho A) \rho B \setminus (B \leftarrow (' \emptyset ' \neq A) v ' ' \neq 1 \phi A) / , A \leftarrow ' ' , \overline{\tau} X$	
159.	Number of digit positions in scalar X (depends on $\square PP$)	$X \leftarrow D0$
	$\rho \overline{\tau} X$	
160.	Leading zeroes for X in fields of width Y	$X \leftarrow I1; Y \leftarrow I0; X \geq 0$
	$\emptyset \ 1 \downarrow (2 \uparrow Y + 1) \overline{\tau} X^{\circ} . + , 10 * Y$	
161.	Row-by-row formatting (width G) of X with Y decimals per row	$X \leftarrow D2; Y \leftarrow I1; G \leftarrow I0$
	$((1, G) \times \rho X) \rho 2 \ 1 \ 3 \phi (\phi G, \rho X) \rho (, G, [1.1] Y) \overline{\tau} \phi X$	
163.	Formatting X with H decimals in fields of width G	$X \leftarrow D; G \leftarrow I1; H \leftarrow I1$
	$(, G, [1.1] H) \overline{\tau} X$	

Roll / Deal ?

164.	Y-shaped array of random numbers within (X[1],X[2])	$X \leftarrow I1; Y \leftarrow I1$
	$X[1] + ?Y \rho - - / X$	
165.	Removing punctuation characters	$X \leftarrow A1$
	$(\sim X \in ' . , : ; ? ' ' ') / X$	
166.	Choosing Y objects out of ιX with replacement (roll)	$Y \leftarrow I; X \leftarrow I$
	$?Y \rho X$	
167.	Choosing Y objects out of ιX without replacement (deal)	$X \leftarrow I0; Y \leftarrow I0$
	$Y ? X$	

Geometrical Functions ○

168.	Arctan $Y \div X$	$X \leftarrow D; Y \leftarrow D$
	$((X \neq 0) \times \sim 3 \circ Y \div X + X = 0) + \circ ((X = 0) \times .5 \times Y) + (X < 0) \times 1 - 2 \times Y < 0$	
169.	Conversion from degrees to radians	$X \leftarrow D$
	$X \times \circ \div 180$	
170.	Conversion from radians to degrees	$X \leftarrow D$
	$X \times 180 \div \circ 1$	
171.	Rotation matrix for angle X (in radians) counter-clockwise	$X \leftarrow D0$
	$2 \ 2 \rho 1 \ \sim 1 \ 1 \ 1 \times 2 \ 1 \ 1 \ 2 \circ X$	

Factorial / Binomial !

172.	Number of permutations of X objects taken Y at a time	$X \leftarrow D; Y \leftarrow D$
	$(!Y) \times Y!X$	
173.	Value of Taylor series with coefficients Y at point X	$X \leftarrow D0; Y \leftarrow D1$
	$+ / Y \times (X^*A) \div !A \leftarrow \bar{1} + \iota \rho Y$	
174.	Poisson distribution of states X with average number Y	$X \leftarrow I; Y \leftarrow D0$
	$(* - Y) \times (Y^*X) \div !X$	
175.	Gamma function	$X \leftarrow D0$
	$!X - 1$	
176.	Binomial distribution of X trials with probability Y	$X \leftarrow I0; Y \leftarrow D0$
	$(A!X) \times (Y^*A) \times (1 - Y)^*X - A \leftarrow -\square I0 - \iota X + 1$	
177.	Beta function	$X \leftarrow D0; Y \leftarrow D0$
	$\div Y \times (X - 1)!Y + X - 1$	
178.	Selecting elements satisfying condition X, others to 1	$X \leftarrow B; Y \leftarrow D$
	$X!Y$	
179.	Number of combinations of X objects taken Y at a time	$X \leftarrow D; Y \leftarrow D$
	$Y!X$	

Index Of ⍷

180.	Removing elements Y from beginning and end of vector X	$X \leftarrow A1; Y \leftarrow A$
	$((A \iota 1) - \square I0) \downarrow (\square I0 - (\phi A \leftarrow \sim X \in Y) \iota 1) \downarrow X$	
181.	Alphabetical comparison with alphabets G	$X \leftarrow A; Y \leftarrow A$
	$(G \iota X) < G \iota Y$	
183.	Sum over elements of X determined by elements of Y	$X \leftarrow D1; Y \leftarrow D1$
	$X + . \times Y \circ . = ((\iota \rho Y) = Y \iota Y) / Y$	
184.	First occurrence of string X in string Y	$X \leftarrow A1; Y \leftarrow A1$
	$(\wedge \neq (\bar{1} + \iota \rho X) \phi X \circ . = Y) \iota 1$	
185.	Removing duplicate rows	$X \leftarrow A2$
	$((A \iota A) = \iota \rho A \leftarrow \square I0 + + \neq \wedge \searrow X v . \neq \phi X) \neq X$	
186.	First occurrence of string X in matrix Y	$X \leftarrow A1; Y \leftarrow A2; \bar{1} \iota \rho Y \leftrightarrow \rho X$
	$(Y \wedge . = X) \iota 1$	
187.	Indices of ones in logical vector X	$X \leftarrow B1$
	$(+ \setminus X) \iota 1 + / X$	
188.	Executing costly monadic function F on repetitive arguments	$X \leftarrow A1$
	$(F \ B / X) [+ \setminus B \leftarrow (X \iota X) = \iota \rho X]$	

189.	Index of (first) maximum element of X	X←D1
	X⋈⌈/X	
190.	Index of first occurrence of elements of Y	X←C1; Y←C1
	⌈/X⋈Y	
191.	Index of (first) minimum element of X	X←D1
	X⋈⌊/X	
192.	Test if each element of X occurs only once	X←A1
	⋈/(X⋈X)=⋈ρX	
193.	Test if all elements of vector X are equal	X←A1
	⋈/⊖I0=X⋈X	
194.	Interpretation of roman numbers	X←A
	+/A× ⁻¹ *A<1⊖A←0,(1000 500 100 50 10 5 1) ['MDCLXVI'⋈X]	
195.	Removing elements Y from end of vector X	X←A1; Y←A
	(⊖I0-(~⊖X∈Y)⋈1)↓X	
196.	Removing trailing blanks	X←C1
	(1-(⊖' '≠X)⋈1)↓X	
198.	Index of last occurrence of Y in X (⊖I0-1 if not found)	X←A1; Y←A
	((⁻¹ 1 1)[2×⊖I0]+ρX)-(⊖X)⋈Y	
199.	Index of last occurrence of Y in X (0 if not found)	X←A1; Y←A
	(1+ρX)-(⊖X)⋈Y	
200.	Index of last occurrence of Y in X, counted from the rear	X←A1; Y←A
	(⊖X)⋈Y	
201.	Index of first occurrence of G in X (circularly) after Y	X←A1; Y←I0; G←A
	⊖I0+(ρX)⋈Y+(Y⊖X)⋈G	
202.	Alphabetizing X; equal alphabets in same column of Y	Y←C2; X←C
	(⁻¹ 1↑ρY)⋈(,Y)⋈X	
203.	Changing index of an unfound element to zero	Y←A1; X←A
	(1+ρY)⋈Y⋈X	
204.	Replacing elements of G in set X with corresponding Y	X←A1, Y←A1, G←A
	A[B/⋈ρB]←Y[(B←B≤ρY)/B←X⋈A←,G] · (ρG)ρA	
205.	Removing duplicate elements (nub)	X←A1
	((X⋈X)=⋈ρX)/X	
206.	First word in X	X←C1
	(⁻¹ 1+X⋈' ')↑X	
207.	Removing elements Y from beginning of vector X	X←A1; Y←A
	(((~X∈Y)⋈1)-⊖I0)↓X	

208.	Removing leading zeroes	$X \leftarrow A1$
	$(\neg 1 + (X = '0') \vee 0) \downarrow X$	
209.	Index of first one after index Y in X	$G \leftarrow I0; X \leftarrow B1$
	$Y + (Y \downarrow X) \vee 1$	
210.	Changing index of an unfound element to zero (not effective)	$X \leftarrow A; Y \leftarrow A1$
	$(X \in Y) \times Y \vee X$	
211.	Indicator of first occurrence of each unique element of X	$X \leftarrow A1$
	$(X \vee X) = \vee \rho X$	
212.	Inverting a permutation	$X \leftarrow I1$
	$X \vee \vee \rho X$	
213.	Index of first differing element in vectors X and Y	$X \leftarrow A1; Y \leftarrow A1$
	$(Y \neq X) \vee 1$	
214.	Which elements of X are not in set Y (difference of sets)	$X \leftarrow A; Y \leftarrow A1$
	$(\neg I0 + \rho Y) = Y \vee X$	
215.	Changing numeric code X into corresponding name in Y	$X \leftarrow D; Y \leftarrow D1; G \leftarrow C2$
	$G[Y \vee X;]$	
216.	Index of key Y in key vector X	$X \leftarrow A1; Y \leftarrow A$
	$X \vee Y$	
217.	Conversion from characters to numeric codes	$X \leftarrow A$
	$\neg AV \vee X$	
218.	Index of first satisfied condition in X	$X \leftarrow B1$
	$X \vee 1$	

Outer Product $\circ . ! \quad \circ . [\quad \circ . |$

219.	Pascal's triangle of order X (binomial coefficients)	$X \leftarrow I0$
	$\neg A \circ . ! A \leftarrow 0, \vee X$	
220.	Maximum table	$X \leftarrow I0$
	$(\vee X) \circ . [\vee X$	
221.	Number of decimals (up to Y) of elements of X	$X \leftarrow D; Y \leftarrow I0$
	$0 + . \neq ([(10 * Y) \times 10 * \neg I0 - \vee Y + 1) \circ . [X \times 10 * Y$	
222.	Greatest common divisor of elements of X	$X \leftarrow I1$
	$[/ (\wedge / 0 = A \circ . X) / A \leftarrow \vee [/ X$	
223.	Divisibility table	$X \leftarrow I1$
	$0 = (\vee [/ X) \circ . X$	
224.	All primes up to X	$X \leftarrow I0$
	$(2 = + \neq 0 = (\vee X) \circ . \vee X) / \vee X$	

Outer Product $\circ$ $\cdot$ $\times$ $\circ$ $\cdot$ $-$ $\circ$ $\cdot$ $+$

225.	Compound interest for principals Y at rates G % in times X	$X \leftarrow D$; $Y \leftarrow D$; $G \leftarrow D$
	$Y \circ \cdot (1 + G \div 100) \circ \cdot X$	
226.	Product of two polynomials with coefficients X and Y	$X \leftarrow D1$; $Y \leftarrow D1$
	$+ \neq (\square I 0 - \iota p X) \phi X \circ \cdot Y, 0 \times 1 \downarrow X$	
228.	Shur product	$X \leftarrow D2$; $Y \leftarrow D2$
	$1 \ 2 \ 1 \ 2 \phi X \circ \cdot Y$	
229.	Direct matrix product	$X \leftarrow D2$; $Y \leftarrow D2$
	$1 \ 3 \ 2 \ 4 \phi X \circ \cdot Y$	
230.	Multiplication table	$X \leftarrow I 0$
	$(\iota X) \circ \cdot \iota X$	
231.	Replicating a dimension of rank three array X Y-fold	$Y \leftarrow I 0$; $X \leftarrow A3$
	$X[; , (Y p 1) \circ \cdot \iota (p X) [2];]$	
232.	Array and its negative ('plus minus')	$X \leftarrow D$
	$X \circ \cdot 1 \ \bar{1}$	
233.	Move set of points X into first quadrant	$X \leftarrow D2$
	$1 \ 2 \ 1 \phi X \circ \cdot - \lfloor / X$	
234.	Test relations of elements of X to range Y; result in $\bar{2}..2$	$X \leftarrow D$; $Y \leftarrow D$; $2 = \bar{1} \uparrow p Y$
	$+ / \times X \circ \cdot - Y$	
235.	Occurrences of string X in string Y	$X \leftarrow A1$; $Y \leftarrow A1$
	$(Y[A \circ \cdot \bar{1} + \iota p X] \wedge . = X) / A \leftarrow (A = 1 \uparrow X) / \iota p A \leftarrow (1 - p X) \downarrow Y$	
236.	Sum of common parts of matrices (matrix sum)	$X \leftarrow D2$; $Y \leftarrow D2$
	$1 \ 2 \ 1 \ 2 \phi X \circ \cdot + Y$	
237.	Adding X to each row of Y	$X \leftarrow D1$; $Y \leftarrow D2$
	$1 \ 1 \ 2 \phi X \circ \cdot + Y$	
238.	Adding X to each row of Y	$X \leftarrow D1$; $Y \leftarrow D2$
	$1 \ 2 \ 1 \phi Y \circ \cdot + X$	
240.	Adding X to each column of Y	$X \leftarrow D1$; $Y \leftarrow D2$
	$2 \ 1 \ 2 \phi X \circ \cdot + Y$	
241.	Adding X to each column of Y	$X \leftarrow D1$; $Y \leftarrow D2$
	$1 \ 2 \ 2 \phi Y \circ \cdot + X$	
242.	Hilbert matrix of order X	$X \leftarrow I 0$
	$\div \bar{1} + (\iota X) \circ \cdot + \iota X$	
243.	Moving index of width Y for vector X	$X \leftarrow A1$; $Y \leftarrow I 0$
	$(0, \iota (p X) - Y) \circ \cdot + Y$	
244.	Indices of subvectors of length Y starting at X+1	$X \leftarrow I1$; $Y \leftarrow I 0$

	$X \circ . + 1Y$	
245.	Reshaping numeric vector X into a one-column matrix	$X \leftarrow D1$
	$X \circ . + , 0$	
246.	Annuity coefficient: X periods at interest rate Y %	$X \leftarrow I; Y \leftarrow D$
	$((pA)pY \div 100) \div A \leftarrow Q1 - (1 + Y \div 100) \circ . * - X$	

Outer Product $\circ . < \circ . \leq \circ . \geq \circ . >$

247.	Matrix with X[i] trailing zeroes on row i	$X \leftarrow I1$
	$X \circ . < \phi 1 \mid / X$	
248.	Matrix with X[i] leading zeroes on row i	$X \leftarrow I1$
	$X \circ . < 1 \mid / X$	
249.	Distribution of X into intervals between Y	$X \leftarrow D; Y \leftarrow D1$
	$+ / ((\neg 1 \downarrow Y) \circ . \leq X) \wedge (1 \downarrow Y) \circ . > X$	
250.	Histogram (distribution barchart; down the page)	$X \leftarrow I1$
	$' \square ' [\square I0 + (\phi 1 \mid / A) \circ . \leq A \leftarrow + / (1 \mid + (\mid / X) - \mid / X) \circ . = X]$	
251.	Barchart of integer values (down the page)	$X \leftarrow I1$
	$' \square ' [\square I0 + (\phi 1 \mid / X) \circ . \leq X]$	
252.	Test if X is an upper triangular matrix	$X \leftarrow D2$
	$\wedge / , (0 \neq X) \leq A \circ . \leq A \leftarrow 1 \uparrow pX$	
253.	Number of ?s intersecting ?s (X=starts, Y=stops)	$X \leftarrow D1; Y \leftarrow D1$
	$+ / A \wedge Q A \leftarrow X \circ . \leq Y$	
254.	Contour levels Y at points with altitudes X	$X \leftarrow D0; Y \leftarrow D1$
	$Y [+ \neq Y \circ . \leq X]$	
255.	X×X upper triangular matrix	$X \leftarrow I0$
	$(1X) \circ . \leq 1X$	
256.	Classification of elements Y into X classes of equal size	$X \leftarrow I0; Y \leftarrow D1$
	$+ / (A \times X \div \mid / A \leftarrow Y - \mid / Y) \circ . \geq \neg 1 + 1X$	
257.	Matrix with X[i] trailing ones on row i	$X \leftarrow I1$
	$X \circ . \geq \phi 1 \mid / X$	
258.	Comparison table	$X \leftarrow I1$
	$X \circ . \geq 1 \mid / X, 0$	
259.	Barchart of X with height Y (across the page)	$X \leftarrow D1; Y \leftarrow D0$
	$' \square ' [\square I0 + X \circ . \geq (\mid / X) \times (1Y) \div Y]$	
260.	Barchart of integer values (across the page)	$X \leftarrow I1$
	$' \square ' [\square I0 + X \circ . \geq 1 \mid / X]$	
261.	Matrix with X[i] leading ones on row i	$X \leftarrow I1$

	$X \circ . \geq \iota \uparrow / X$	
263.	Test if X is a lower triangular matrix $\wedge / , (0 \neq X) \leq A \circ . \geq A \leftarrow \iota \uparrow \rho X$	$X \leftarrow D2$
264.	Test if X is within range [Y[1],Y[2]) $\neq / X \circ . \geq Y$	$X \leftarrow D; Y \leftarrow D1$
265.	Ordinal numbers of words in X that indices Y point to $\square I0++/Y \circ . \geq (' ' = X) / \iota \rho X$	$X \leftarrow C1; Y \leftarrow I$
266.	Which class do elements of X belong to $+ / X \circ . \geq 0 \ 50 \ 100 \ 1000$	$X \leftarrow D$
267.	$X \times X$ lower triangular matrix $(\iota X) \circ . \geq \iota X$	$X \leftarrow I0$
268.	Moving all blanks to end of each row $(\rho X) \rho (, (+ / A) \circ . > - \square I0 - \iota \uparrow \rho X) \setminus (, A \leftarrow X \neq ' ') / , X$	$X \leftarrow C$
269.	Justifying right fields of X (lengths Y) to length G $(, Y \circ . > \phi (\iota G) - \square I0) \setminus X$	$X \leftarrow A1; Y \leftarrow I1; G \leftarrow I0$
270.	Justifying left fields of X (lengths Y) to length G $(, Y \circ . > (\iota G) - \square I0) \setminus X$	$X \leftarrow A1; Y \leftarrow I1; G \leftarrow I0$

Outer Product $\circ . \neq$ $\circ . =$

271.	Indices of elements of Y in corr. rows of X ($X[i;] \iota Y[i;]$) $1++/\wedge \setminus 1 \ 2 \ 1 \ 3 \phi Y \circ . \neq X$	$X \leftarrow A2; Y \leftarrow A2$
273.	Indicating equal elements of X as a logical matrix $\phi X \circ . = (1 \ 1 \phi < \setminus X \circ . = X) / X$	$X \leftarrow A1$
275.	Changing connection matrix X ($\uparrow 1 \rightarrow 1$) to a node matrix $(1 \uparrow 1 \circ . = \phi X) + . \times \iota \uparrow \rho \square \leftarrow X$	$X \leftarrow I2$
276.	Sums according to codes G $(G \circ . = X) + . \times Y$	$X \leftarrow A; Y \leftarrow D; G \leftarrow A$
277.	Removing duplicate elements (nub) $(1 \ 1 \phi < \setminus X \circ . = X) / X$	$X \leftarrow A1$
278.	Changing node matrix X (starts,ends) to a connection matrix $- / (\iota \uparrow / , X) \circ . = \phi X$	$X \leftarrow I2$
279.	Test if all elements of vector X are equal $v / \wedge / 0 \ 1 \circ . = X$	$X \leftarrow B1$
280.	Test if elements of X belong to corr. row of Y ($X[i;] \in Y[i;]$) $v / 1 \ 2 \ 1 \ 3 \phi X \circ . = Y$	$X \leftarrow A2; Y \leftarrow A2; 1 \uparrow \rho X \leftrightarrow 1 \uparrow \rho Y$
281.	Test if X is a permutation vector	$X \leftarrow I1$

	$\Lambda/1=+\neq X\circ.=\iota\rho X$	
282.	Occurrences of string X in string Y $(\Lambda\neq(\neg 1+\iota\rho X)\phi(X\circ.=Y),\theta)/\iota 1+\rho Y$	$X\leftarrow C1; Y\leftarrow C1$
283.	Division to Y classes with width H, minimum G $+/(\iota Y)\circ.=\lceil (X-G)\div H$	$X\leftarrow D; Y\leftarrow I0; G\leftarrow D0; H\leftarrow D0$
285.	Repeat matrix $((\neg 1\phi\sim A)\wedge A\leftarrow(\neg 1\downarrow X=1\phi X),\theta)/Y)\circ.=Y$	$X\leftarrow A1; Y\leftarrow A1$
286.	X×X identity matrix $(\iota X)\circ.=\iota X$	$X\leftarrow I0$

Inner Product $\lceil.\times \lfloor.\times \lceil.+ \times.\circ \times.* +.*$

287.	Maxima of elements of subsets of X specified by Y $A+(X-A\leftarrow\lfloor/X)\lceil.\times Y$	$X\leftarrow A1; Y\leftarrow B$
288.	Indices of last non-blanks in rows $(\neg ' '\neq X)\lceil.\times \iota \neg 1\uparrow\rho X$	$X\leftarrow C$
289.	Maximum of X with weights Y $Y\lceil.\times X$	$X\leftarrow D1; Y\leftarrow D1$
290.	Minimum of X with weights Y $Y\lfloor.\times X$	$X\leftarrow D1; Y\leftarrow D1$
292.	Extending a distance table to next leg $X\leftarrow X\lfloor.+X$	$X\leftarrow D2$
293.	A way to combine trigonometric functions (sin X cos Y) $1\ 2\times.\circ X,Y$	$X\leftarrow D0; Y\leftarrow D0$
294.	Sine of a complex number $(2\ 2\rho 1\ 6\ 2\ 5)\times.\circ X$	$X\leftarrow D; 2=1\uparrow\rho X$
295.	Products over subsets of X specified by Y $X\times.*Y$	$X\leftarrow A1; Y\leftarrow B$
296.	Sum of squares of X $X+.*2$	$X\leftarrow D1$
297.	Randomizing random numbers (in $\square LX$ in a workspace) $\square RL\leftarrow\square TS+.*2$	

Inner Product $V.\wedge <.< <.\leq <.\geq \leq.\geq >.>$

298.	Extending a transitive binary relation $X\leftarrow XV.V.\wedge X$	$X\leftarrow B2$
------	---	------------------

299.	Test if X is within range [Y[1;],Y[2;])	X←D0; Y←D2; 1↑pY ↔ 2
	X<.<Y	
300.	Test if X is within range (Y[1;],Y[2;]]	X←D0; Y←D2; 1↑pY ↔ 2
	X<.<=Y	
301.	Test if X is within range (Y[1;],Y[2;]]	X←D; Y←D2; 1↑pY ↔ 2
	X<.<=Y	
302.	Test if the elements of X are ascending	X←D1
	X<.>1ϕX	
303.	Test if X is an integer within range [G,H)	X←I0; G←I0; H←I0
	~X≤.>([X],G,H	
304.	Test if X is within range (Y[1;],Y[2;]]	X←D; Y←D2; 1↑pY ↔ 2
	(X,[.1+ppX]X)>.>Y	

Inner Product v.≠ Λ.= +.≠ +.=

306.	Removing trailing blank columns	X←C2
	(ϕv\ϕ' 'v.≠X)/X	
307.	Removing leading blank rows	X←C2
	(v\Xv.≠' ')≠X	
308.	Removing leading blank columns	X←C2
	(v\ ' 'v.≠X)/X	
309.	Index of first occurrences of rows of X as rows of Y	X←A, Y←A2
	⊖I0++≠Λ≠Yv.≠X	
310.	X∩Y for rows of matrices	X←A2; Y←A2
	⊖I0++≠Λ≠Xv.≠Y	
311.	Removing duplicate blank rows	X←C2
	(Av1↓1ϕ1,A←Xv.≠' ')≠X	
312.	Removing duplicate blank columns	X←C2
	(Av1,~1↓A←' 'v.≠X)/X	
313.	Removing blank columns	X←C2
	(' 'v.≠X)/X	
314.	Removing blank rows	X←C2
	(Xv.≠' ')≠X	
315.	Test if rows of X contain elements differing from Y	X←A; Y←A0
	Xv.≠Y	
316.	Removing trailing blank rows	X←C2
	(-2↑+/\ϕXΛ.= ' ')↓X	

317.	Removing duplicate rows	$X \leftarrow A2$
	$(v \neq \setminus X \wedge . = \emptyset X) \neq X$	
318.	Removing duplicate rows	$X \leftarrow A2$
	$(1 \setminus \emptyset < \setminus X \wedge . = \emptyset X) \neq X$	
319.	Test if circular lists are equal (excluding phase)	$X \leftarrow A1; Y \leftarrow A1$
	$v / Y \wedge . = \emptyset (\setminus \rho X) \phi (2 \rho \rho X) \rho X$	
320.	Test if all elements of vector X are equal	$X \leftarrow B1$
	$X \wedge . = v / X$	
321.	Test if all elements of vector X are equal	$X \leftarrow B1$
	$X \wedge . = \wedge / X$	
322.	Rows of matrix X starting with string Y	$X \leftarrow A2; Y \leftarrow A1$
	$((((1 \uparrow \rho X), \rho Y) \uparrow X) \wedge . = Y) \neq X$	
323.	Occurrences of string X in string Y	$X \leftarrow A1; Y \leftarrow A1$
	$((-A) \downarrow X \wedge . = (A, 1 + \rho Y) \rho Y) / \setminus (\rho Y) + 1 - A \leftarrow \rho X$	
324.	Test if vector Y is a row of array X	$X \leftarrow A; Y \leftarrow A1$
	$1 \in X \wedge . = Y$	
325.	Comparing vector Y with rows of array X	$X \leftarrow A; Y \leftarrow A1$
	$X \wedge . = Y$	
326.	Word lengths of words in list X	$X \leftarrow C$
	$X + . \neq ' '$	
327.	Number of occurrences of scalar X in array Y	$X \leftarrow A0; Y \leftarrow A$
	$X + . = , Y$	
328.	Counting pairwise matches (equal elements) in two vectors	$X \leftarrow A1; Y \leftarrow A1$
	$X + . = Y$	

Inner Product $- . \div + . \div + . \times$

329.	Sum of alternating reciprocal series $Y \div X$	$X \leftarrow D1; Y \leftarrow D1$
	$Y - . \div X$	
330.	Limits X to fit in $\mathbb{F}$ field $Y[1\ 2]$	$X \leftarrow D; Y \leftarrow I1$
	$(X[1 \downarrow A][1 \uparrow A \leftarrow (2 \setminus \rho^{-1} \ 1 \ 1 \setminus .1) + . \times 10^*(-1 \downarrow Y), - / Y + Y > 99 \ 0$	
331.	Value of polynomial with coefficients Y at point X	$X \leftarrow D0; Y \leftarrow D$
	$(X^{\setminus 1} + \setminus \rho Y) + . \times \phi Y$	
332.	Arithmetic average (mean value) of X weighted by Y	$X \leftarrow D1; Y \leftarrow D1$
	$(Y + . \times X) \div \rho X$	
333.	Scalar (dot) product of vectors	$X \leftarrow D1; Y \leftarrow D1$
	$Y + . \times X$	

334.	Sum of squares of X	$X \leftarrow D1$
	$X +. \times X$	
335.	Summation over subsets of X specified by Y	$X \leftarrow A1; Y \leftarrow B$
	$X +. \times Y$	
336.	Matrix product	$X \leftarrow D; Y \leftarrow D; \neg 1 \uparrow \rho X \leftrightarrow 1 \uparrow \rho Y$
	$X +. \times Y$	
337.	Sum of reciprocal series $Y \div X$	$X \leftarrow D1; Y \leftarrow D1$
	$Y +. \div X$	

Scan $\backslash$ $\backslash$ $\times \backslash$ $- \backslash$

338.	Groups of ones in Y pointed to by X (or trailing parts)	$X \leftarrow B; Y \leftarrow B$
	$Y \wedge A = [\backslash X \times A \leftarrow + \backslash Y > \neg 1 \downarrow 0, Y$	
339.	Test if X is in ascending order along direction Y	$X \leftarrow D; Y \leftarrow I0$
	$\wedge / [Y] X = [\backslash [Y] X$	
340.	Duplicating element of X belonging to Y, $1 \uparrow X$ until next found	$X \leftarrow A1; Y \leftarrow B1$
	$X[1][\backslash Y \times \neg 1 \rho Y]$	
341.	Test if X is in descending order along direction Y	$X \leftarrow D; Y \leftarrow I0$
	$\wedge / [Y] X = [\backslash [Y] X$	
342.	Value of Taylor series with coefficients Y at point X	$X \leftarrow D0; Y \leftarrow D1$
	$+ / Y \times \times \neg 1, X \div \neg 1 + \rho Y$	
343.	Alternating series ($1 \neg 1 2 \neg 2 3 \neg 3 \dots$)	$X \leftarrow I0$
	$- \neg 1 X$	

Scan $\neq \backslash$ $< \backslash$ $\leq \backslash$ $\neq \backslash$

346.	Value of saddle point	$X \leftarrow D2$
	$(< \backslash, (X = (\rho X) \rho [\neq X] \wedge X = \Phi(\Phi \rho X) \rho [/X]) / , X$	
348.	First one (turn off all ones after first one)	$X \leftarrow B$
	$< \backslash X$	
350.	Not first zero (turn on all zeroes after first zero)	$X \leftarrow B$
	$\leq \backslash X$	
351.	Running parity ($\neq \backslash$) over subvectors of Y indicated by X	$X \leftarrow B1; Y \leftarrow B1$
	$\neq \backslash Y \neq X \backslash A \neq \neg 1 \downarrow 0, A \leftarrow X / \neq \neg 1 \downarrow 0, Y$	
352.	Vector $(X[1] \rho 1), (X[2] \rho 0), (X[3] \rho 1), \dots$	$X \leftarrow I1$
	$\neq \backslash (\neg 1 / X) \in + \backslash \square I0, X$	
353.	Not leading zeroes($\vee \backslash$) in each subvector of Y indicated by X	$X \leftarrow B1; Y \leftarrow B1$

	$\neq \backslash (Y \vee X) \backslash A \neq^{-1} \downarrow 0, A \leftarrow (Y \vee X) / Y$	
354.	Leading ones ($\wedge \backslash$) in each subvector of Y indicated by X	$X \leftarrow B1; Y \leftarrow B1$
	$\sim \neq \backslash (Y \leq X) \backslash A \neq^{-1} \downarrow 0, A \leftarrow \sim (Y \leq X) / Y$	
355.	Locations of texts between and including quotes	$X \leftarrow C1$
	$A \vee^{-1} \downarrow 0, A \leftarrow \neq \backslash X = ' ' ' '$	
356.	Locations of texts between quotes	$X \leftarrow C1$
	$A \wedge^{-1} \downarrow 0, A \leftarrow \neq \backslash X = ' ' ' '$	
357.	Joining pairs of ones	$X \leftarrow B$
	$X \vee \neq \backslash X$	
358.	Places between pairs of ones	$X \leftarrow B$
	$(\sim X) \wedge \neq \backslash X$	
359.	Running parity	$X \leftarrow B$
	$\neq \backslash X$	

Scan $\vee \backslash \wedge \backslash$

360.	Removing leading and trailing blanks	$X \leftarrow C1$
	$((\phi \vee \backslash \phi A) \wedge \vee \backslash A \leftarrow ' ' \neq X) / X$	
361.	First group of ones	$X \leftarrow B$
	$X \wedge \wedge \backslash X = \vee \backslash X$	
362.	Removing trailing blank columns	$X \leftarrow C2$
	$(\phi \vee \backslash \phi \vee \neq ' ' \neq X) / X$	
363.	Removing trailing blanks	$X \leftarrow C1$
	$(\phi \vee \backslash \phi ' ' \neq X) / X$	
364.	Removing leading blanks	$X \leftarrow C1$
	$(\vee \backslash ' ' \neq X) / X$	
365.	Not leading zeroes (turn on all zeroes after first one)	$X \leftarrow B$
	$\vee \backslash X$	
366.	Centering character array X with ragged edges	$X \leftarrow C$
	$(A - \lfloor 0.5 \times (A \leftarrow + / \wedge \backslash \phi A) ++ / \wedge \backslash A \leftarrow ' ' = \phi X) \phi X$	
367.	Decommenting a matrix representation of a function ($\square CR$)	$X \leftarrow C2$
	$(\vee / A) \neq (\rho X) \rho (, A) \backslash (, A \leftarrow \wedge \backslash (' \# ' \neq X) \vee \neq \backslash X = ' ' ' ') / , X$	
369.	Centering character array X with only right edge ragged	$X \leftarrow C$
	$(- \lfloor 0.5 \times + / \wedge \backslash ' ' = \phi X) \phi X$	
370.	Justifying right	$X \leftarrow C$
	$(- + / \wedge \backslash \phi ' ' = X) \phi X$	
371.	Removing trailing blanks	$X \leftarrow C1$

	$(-+/\wedge\phi' '=X)\downarrow X$	
372.	Justifying left $(+/\wedge\backslash' '=X)\phi X$	$X\leftarrow C$
373.	Editing X with Y '-wise $((\sim(\rho A\uparrow X)\uparrow' /'=Y)/A\uparrow X), (1\downarrow A\downarrow Y), (A\leftarrow +/\wedge\backslash Y\neq', ') \downarrow X$	$X\leftarrow C1; Y\leftarrow C1$
374.	Removing leading blanks $(+/\wedge\backslash' '=X)\downarrow X$	$X\leftarrow C1$
375.	Indices of first blanks in rows of array X $\square I0++/\wedge\backslash' '\neq X$	$X\leftarrow C$
377.	Leading ones (turn off all ones after first zero) $\wedge\backslash X$	$X\leftarrow B$

Scan +\

378.	Vector $(X[1]\rho 1), (Y[1]\rho 0), (X[2]\rho 1), \dots$ $(\iota+/X, Y) \in +\backslash 1+\backslash 1\downarrow 0, ((\iota+/X) \in +\backslash X) \backslash Y$	$X\leftarrow I1; Y\leftarrow I1$
379.	Replicate Y[i] X[i] times (for all i) $((X\neq 0)/Y) [+ \backslash 1\phi(\iota+/X) \in +\backslash X]$	$X\leftarrow I1; Y\leftarrow A1$
380.	Vector $(Y[1]+\iota X[1]), (Y[2]+\iota X[2]), (Y[3]+\iota X[3]), \dots$ $\square I0++\backslash 1+((\iota+/X) \in +\backslash \square I0, X) \backslash Y-\backslash 1\downarrow 1, X+Y$	$X\leftarrow I1; Y\leftarrow I1; \rho X\leftrightarrow \rho Y$
381.	Replicate Y[i] X[i] times (for all i) $Y[+\backslash(\iota+/X) \in \backslash 1\downarrow 1++\backslash 0, X]$	$X\leftarrow I1; Y\leftarrow A1$
382.	Replicate Y[i] X[i] times (for all i) $Y[\square I0++\backslash(\iota+/X) \in \square I0++\backslash X]$	$X\leftarrow I1; Y\leftarrow A1$
383.	Cumulative sums (+) over subvectors of Y indicated by X $+\backslash Y-X\backslash A-\backslash 1\downarrow 0, A\leftarrow X/+\backslash 1\downarrow 0, Y$	$X\leftarrow B1; Y\leftarrow D1$
384.	Sums over (+) subvectors of Y, lengths in X $A-\backslash 1\downarrow 0, A\leftarrow (+\backslash Y) [+ \backslash X]$	$X\leftarrow I1; Y\leftarrow D1$
386.	X first figurate numbers $+\backslash +\backslash \iota X$	$X\leftarrow I0$
387.	Insert vector for X[i] zeroes after i:th subvector $(\iota(\rho Y)++/X) \in +\backslash 1+\backslash 1\downarrow 0, (1\phi Y) \backslash X$	$X\leftarrow I1; Y\leftarrow B1$
388.	Open a gap of X[i] after Y[G[i]] (for all i) $((\iota(\rho Y)++/X) \in +\backslash 1+\backslash 1\downarrow 0, ((\iota\rho Y) \in G) \backslash X) \backslash Y$	$X\leftarrow I1; Y\leftarrow A1; G\leftarrow I1$
389.	Open a gap of X[i] before Y[G[i]] (for all i) $((\iota(\rho Y)++/X) \in +\backslash 1+((\iota\rho Y) \in G) \backslash X) \backslash Y$	$X\leftarrow I1; Y\leftarrow A1; G\leftarrow I1$
390.	Changing lengths X of subvectors to starting indicators	$X\leftarrow I1$

26/46

	$(\iota pY) \in + \backslash \square I0, X$	
409.	X first triangular numbers	$X \leftarrow I0$
	$+ \backslash \iota X$	
410.	Cumulative sum	$X \leftarrow D$
	$+ \backslash X$	

Reduction $\circ/$ $\div/$ $-/$ $\times/$

411.	Complementary angle (arccos sin X)	$X \leftarrow D0$
	$\circ/ \sim^2 1, X$	
412.	Evaluating a two-row determinant	$X \leftarrow D2$
	$- / \times / 0 \ 1 \ominus X$	
413.	Evaluating a two-row determinant	$X \leftarrow D2$
	$- / \times \neq 0 \ 1 \phi X$	
414.	Area of triangle with side lengths in X (Heron's formula)	$X \leftarrow D1; \ 3 \leftrightarrow \rho X$
	$(\times / (+ / X \div 2) - 0, X) * .5$	
415.	Juxtapositioning planes of rank 3 array X	$X \leftarrow A3$
	$(\times \neq 2 \ 2 \rho 1, \rho X) \rho 2 \ 1 \ 3 \phi X$	
416.	Number of rows in array X (also of a vector)	$X \leftarrow A$
	$\times / \sim^1 \downarrow \rho X$	
417.	(Real) solution of quadratic equation with coefficients X	$X \leftarrow D1; \ 3 \leftrightarrow \rho X$
	$(-X[2] - \sim^1 1 \times ((X[2]*2) - \times / 4, X[1 \ 3])) * .5 \div 2 \times X[1]$	
418.	Reshaping planes of rank 3 array to rows of a matrix	$X \leftarrow A3$
	$(\times / 2 \ 2 \rho 1, \rho X) \rho X$	
419.	Reshaping planes of rank 3 array to a matrix	$X \leftarrow A3$
	$(\times / 2 \ 2 \rho (\rho X), 1) \rho X$	
420.	Number of elements (also of a scalar)	$X \leftarrow A$
	$\times / \rho X$	
421.	Product of elements of X	$X \leftarrow D1$
	$\times / X$	
422.	Alternating product	$X \leftarrow D$
	$\div / X$	
423.	Centering text line X into a field of width Y	$X \leftarrow C1; \ Y \leftarrow I0$
	$Y \uparrow ((\lfloor - / .5 \times Y, \rho X) \rho ' \ '), X$	
424.	Alternating sum	$X \leftarrow D$
	$- / X$	

Reduction $\lceil /$ $\lfloor /$

425.	Test if all elements of vector X are equal	X←D1
	$(\lceil /X) = \lfloor /X$	
426.	Size of range of elements of X	X←D1
	$(\lceil /X) - \lfloor /X$	
427.	Conversion of set of positive integers X to a mask	X←I1
	$(\lceil \lceil /X) \in X$	
428.	Negative infinity; the smallest representable value	
	$\lceil / \lceil 0$	
429.	Vectors as column matrices in catenation beneath each other	X←A1/2; Y←A1/2
	$X, [1+.5 \times \lceil /(\rho\rho X), \rho\rho Y]Y$	
430.	Vectors as row matrices in catenation upon each other	X←A1/2; Y←A1/2
	$X, [.5 \times \lceil /(\rho\rho X), \rho\rho Y]Y$	
431.	Quick membership ($\in$) for positive integers	X←I1; Y←I1
	$A \leftarrow (\lceil /X, Y) \rho 0 \cdot A[Y] \leftarrow 1 \cdot A[X]$	
432.	Positive maximum, at least zero (also for empty X)	X←D1
	$\lceil /X, 0$	
433.	Maximum of elements of X	X←D1
	$\lceil /X$	
434.	Positive infinity; the largest representable value	
	$\lceil / \lceil 0$	
435.	Minimum of elements of X	X←D1
	$\lfloor /X$	

Reduction $\vee /$ $\wedge /$ $\neq /$

436.	Test if all elements of vector X are equal	X←B1
	$\wedge / 0 \ 1 \in X$	
437.	Test if all elements of vector X are equal	X←B1
	$(\wedge /X) \vee \sim \vee /X$	
438.	Test if all elements of vector X are equal	X←B1
	$(\wedge /X) = \vee /X$	
439.	Test if all elements of vector X are equal	X←B1
	$\wedge /X \div \vee /X$	
440.	Removing duplicate rows from ordered matrix X	X←A2
	$(\sim 1 \phi 1 \downarrow (\vee /X \neq 1 \ominus X), 1) \nearrow X$	

441.	Vector having as many ones as X has rows	X←A2
	v/0/X	
442.	Test if X and Y have elements in common	X←A; Y←A1
	v/Y∈X	
443.	None, neither	X←B
	~v/X	
444.	Any, anyone	X←B
	v/X	
445.	Test if all elements of vector X are equal	X←B1
	≠/0 1∈X	
446.	Parity	X←B
	≠/X	

Reduction $\wedge$ /

447.	Number of areas intersecting areas in X	X←D3 (n × 2 × dim)
	+/A∧⊖A←∧/X[;Ap1;]≤2 1 3⊖X[; (A←1↑pX)p2;]	
448.	Test if all elements of vector X are equal	X←B1
	∧/X/1⊖X	
449.	Comparison of successive rows	X←A2
	∧/X=1⊖X	
450.	Test if all elements of vector X are equal	X←A1
	∧/X=1⊖X	
451.	Test if X is a valid APL name	X←C1
	∧/((1↑X)∈10↓A),X∈A←'0..9A..Za..z'	
452.	Test if all elements of vector X are equal	X←A1
	∧/X=1↑X	
453.	Identity of two sets	X←A1; Y←A1
	∧/(X∈Y),Y∈X	
454.	Test if X is a permutation vector	X←I1
	∧/(1ρX)∈X	
455.	Test if all elements of vector X are equal	X←B1
	~∧/X∈~X	
456.	Test if X is boolean	X←A
	∧/,X∈0 1	
457.	Test if Y is a subset of X (Y ⊂ X)	X←A; Y←A1
	∧/Y∈X	

458.	Test if arrays of equal shape are identical	$X \leftarrow A; Y \leftarrow A; \rho X \leftrightarrow \rho Y$
	$\wedge /, X=Y$	
459.	Test if all elements of vector X are equal	$X \leftarrow A1$
	$\wedge / X=X[1]$	
460.	Blank rows	$X \leftarrow C2$
	$\wedge / ' '=X$	
461.	All, both	$X \leftarrow B$
	$\wedge / X$	

Reduction +/

462.	Standard deviation of X	$X \leftarrow D1$
	$((+/ (X - (+/X) \div \rho X) * 2) \div \rho X) * .5$	
463.	Y:th moment of X	$X \leftarrow D1$
	$(+/ (X - (+/X) \div \rho X) * Y) \div \rho X$	
464.	Variance (dispersion) of X	$X \leftarrow D1$
	$(+/ (X - (+/X) \div \rho X) * 2) \div \rho X$	
465.	Arithmetic average (mean value), also for an empty array	$X \leftarrow D$
	$(+/ , X) \div 1 \uparrow \rho , X$	
466.	Test if all elements of vector X are equal	$X \leftarrow B1$
	$0 = (\rho X) +/X$	
467.	Average (mean value) of columns of matrix X	$X \leftarrow D2$
	$(+ \nearrow X) \div 1 \uparrow (\rho X) , 1$	
468.	Average (mean value) of rows of matrix X	$X \leftarrow D2$
	$(+/X) \div ^\sim 1 \uparrow 1 , \rho X$	
469.	Number of occurrences of scalar X in array Y	$X \leftarrow A0; Y \leftarrow A$
	$+ / X = , Y$	
470.	Average (mean value) of elements of X along direction Y	$X \leftarrow D; Y \leftarrow I0$
	$(+/ [Y] X) \div (\rho X) [Y]$	
471.	Arithmetic average (mean value)	$X \leftarrow D1$
	$(+/X) \div \rho X$	
472.	Resistance of parallel resistors	$X \leftarrow D1$
	$\div + / \div X$	
473.	Sum of elements of X	$X \leftarrow D1$
	$+ / X$	
474.	Row sum of a matrix	$X \leftarrow D2$
	$+ / X$	

475.	Column sum of a matrix	$X \leftarrow D2$
	$+\neq X$	
476.	Reshaping one-element vector X into a scalar	$X \leftarrow A1$
	$+/X$	
477.	Number of elements satisfying condition X	$X \leftarrow B1$
	$+/X$	

Reverse ϕ Θ

478.	Scan from end with function α	$X \leftarrow A$
	$\phi\alpha\backslash\phi X$	
479.	The index of positive integers in Y	$X \leftarrow I; Y \leftarrow I1$
	$A \leftarrow 99999\rho\lceil I0 + \rho Y \cdot A[\phi Y] \leftarrow \phi\lceil\rho Y \cdot A[X]$	
480.	'Transpose' of matrix X with column fields of width Y	$X \leftarrow A2; G \leftarrow I0$
	$((\phi A) \times 1, Y) \rho 2 \ 1 \ 3 \phi(1\phi Y, A \leftarrow (\rho X) \div 1, Y) \rho X$	
482.	Adding X to each column of Y	$X \leftarrow D1; Y \leftarrow D; (\rho X) = 1 \uparrow \rho Y$
	$Y + \phi(\phi\rho Y) \rho X$	
483.	Matrix with shape of Y and X as its columns	$X \leftarrow A1; Y \leftarrow A2$
	$\phi(\phi\rho Y) \rho X$	
484.	Derivate of polynomial X	$X \leftarrow D1$
	$\lceil 1 \downarrow X \times \phi \lceil 1 + \lceil \rho X$	
485.	Reverse vector X on condition Y	$X \leftarrow A1; Y \leftarrow B0$
	$, \phi[\lceil I0 + Y](1, \rho X) \rho X$	
486.	Reshaping vector X into a one-column matrix	$X \leftarrow A1$
	$(\phi 1, \rho X) \rho X$	
487.	Avoiding parentheses with help of reversal	
	$(\phi 1, \dots)$	

Rotate ϕ Θ

488.	Vector (cross) product of vectors	$X \leftarrow D; Y \leftarrow D$
	$((1\phi X) \times \lceil 1\phi Y) - (\lceil 1\phi X) \times 1\phi Y$	
489.	A magic square, side X	$X \leftarrow I0; 1 = 2 \mid X$
	$A \Theta (A \leftarrow (\lceil X) - \lceil X \div 2) \phi(X, X) \rho \lceil X \times X$	
490.	Removing duplicates from an ordered vector	$X \leftarrow A1$
	$(\lceil 1\phi 1 \downarrow (X \neq \lceil 1\phi X), 1) / X$	
491.	<u>An expression giving itself</u>	

	$1\phi 22\rho 11\rho'''1\phi 22\rho 11\rho'''$	
492.	Transpose matrix X on condition Y $(Y\phi 1\ 2)\Phi X$	$X\leftarrow A2; Y\leftarrow B0$
493.	Any element true ($\vee$ /) on each subvector of Y indicated by X $(X/Y)\geq A/1\phi A\leftarrow(Y\vee X)/X$	$X\leftarrow B1; Y\leftarrow B1$
494.	All elements true ($\wedge$ /) on each subvector of Y indicated by X $(X/Y)\wedge A/1\phi A\leftarrow(Y\leq X)/X$	$X\leftarrow B1; Y\leftarrow B1$
495.	Removing leading, multiple and trailing Y's $(1\uparrow A)\downarrow(A\wedge 1\phi A\leftarrow Y=X)/X$	$X\leftarrow A1; Y\leftarrow A0$
496.	Changing starting indicators X of subvectors to lengths $A-\neg 1\downarrow 0, A\leftarrow(1\phi X)/\uparrow\rho X$	$X\leftarrow B1$
498.	(Cyclic) compression of successive blanks $(A\vee 1\phi A\leftarrow X\neq''')/X$	$X\leftarrow C1$
499.	Aligning columns of matrix X to diagonals $(1-\uparrow\neg 1\uparrow\rho X)\phi X$	$X\leftarrow A2$
500.	Aligning diagonals of matrix X to columns $(\neg 1+\uparrow\neg 1\uparrow\rho X)\phi X$	$X\leftarrow A2$
501.	Diagonal matrix with elements of X $0-\neg 1\downarrow(-\uparrow\rho X)\phi((2\rho\rho X)\rho 0), X$	$X\leftarrow D1$
502.	Test if elements differ from previous ones (non-empty X) $1, 1\downarrow X\neq\neg 1\phi X$	$X\leftarrow A1$
503.	Test if elements differ from next ones (non-empty X) $(\neg 1\downarrow X\neq 1\phi X), 1$	$X\leftarrow A1$
504.	Replacing first element of X with Y $\neg 1\phi 1\downarrow X, Y$	$X\leftarrow A1; Y\leftarrow A0$
505.	Replacing last element of X with Y $1\phi\neg 1\downarrow Y, X$	$X\leftarrow A1; Y\leftarrow A0$
506.	Ending points for X in indices pointed by Y $1\phi(\uparrow\rho X)\in Y$	$X\leftarrow A1; Y\leftarrow I1$
507.	Leftmost neighboring elements cyclically $\neg 1\phi X$	$X\leftarrow A$
508.	Rightmost neighboring elements cyclically $1\phi X$	$X\leftarrow A$

Transpose Φ

509.	Applying to columns action defined on rows	$X\leftarrow A1; Y\leftarrow I0$
------	--	----------------------------------

	$\Phi \dots \Phi X$	
510.	Retrieving scattered elements Y from matrix X $1 \ 1\Phi X[Y[1;];Y[2;]]$	$X \leftarrow A2; Y \leftarrow I2$
511.	Successive transposes of G (X after Y: $X\Phi Y\Phi G$) $X[Y]\Phi G$	$X \leftarrow I1; Y \leftarrow I1$
512.	Major diagonal of array X $(1 * \rho X) \Phi X$	$X \leftarrow A$
513.	Reshaping a 400×12 character matrix to fit into one page $40 \ 120\rho 2 \ 1 \ 3\Phi 10 \ 40 \ 12\rho X$	$X \leftarrow C2$
514.	Transpose of planes of a rank three array $1 \ 3 \ 2\Phi X$	$X \leftarrow A3$
515.	Major diagonal of matrix X $1 \ 1\Phi X$	$X \leftarrow A2$
516.	Selecting specific elements from a 'large' outer product $G\Phi X \circ . \alpha Y$	$X \leftarrow A; Y \leftarrow A; G \leftarrow I1$
517.	Test for antisymmetry of square matrix X $\sim 0 \in X = -\Phi X$	$X \leftarrow D2$
518.	Test for symmetry of square matrix X $\sim 0 \in X = \Phi X$	$X \leftarrow A2$
519.	Matrix with X columns Y $\Phi(X, \rho Y) \rho Y$	$X \leftarrow I0; Y \leftarrow D1$

Maximum [Minimum]

520.	Limiting X between Y[1] and Y[2], inclusive $Y[1][Y[2]] \downarrow X$	$X \leftarrow D; Y \leftarrow D1$
521.	Inserting vector Y to the end of matrix X $(A \uparrow X), [\downarrow 1] (1 \downarrow A \leftarrow (\rho X) [0, \rho Y) \uparrow Y$	$X \leftarrow A2; Y \leftarrow A1$
522.	Widening matrix X to be compatible with Y $((0 \ 1 \times \rho Y) [\rho X) \uparrow X$	$X \leftarrow A2; Y \leftarrow A2$
523.	Lengthening matrix X to be compatible with Y $((1 \ 0 \times \rho Y) [\rho X) \uparrow X$	$X \leftarrow A2; Y \leftarrow A2$
524.	Reshaping non-empty lower-rank array X into a matrix $(1 [\sim 2 \uparrow \rho X) \rho X$	$X \leftarrow A; 2 \geq \rho \rho X$
525.	Take of at most X elements from Y $(X [\rho Y) \uparrow Y$	$X \leftarrow I; Y \leftarrow A$
526.	Limiting indices and giving a default value G $X \leftarrow A1; Y \leftarrow I; G \leftarrow A0$	

(X,G)[(1+pX)⌊Y]

Ceiling ⌈ Floor ⌋

527.	Reshaping X into a matrix of width Y	X←D, Y←I0
	$((\lceil (p,X) \div Y), Y) pX$	
528.	Rounding to nearest even integer	X←D
	$\lfloor X+1 \leq 2 \rfloor X$	
529.	Rounding, to nearest even integer for .5 = 1 ⌊ X	X←D
	$\lfloor X+.5 \times .5 \neq 2 \rfloor X$	
530.	Rounding, to nearest even integer for .5 = 1 ⌊ X	X←D
	$\lfloor X+.5 \times .5 \neq 2 \rfloor X$	
531.	Arithmetic progression from X to Y with step G	X←D0; Y←D0; G←D0
	$X+(G \times Y-X) \times (\lceil 1+\lfloor (Y-X) \div G \rfloor - \square I0$	
532.	Centering text line X into a field of width Y	X←C1; Y←I0
	$(-\lfloor .5 \times Y+pX \rfloor) \uparrow X$	
533.	Test if integer	X←D
	$X=\lfloor X$	
534.	Rounding currencies to nearest 5 subunits	X←D
	$.05 \times \lfloor .5+X \div .05$	
535.	First part of numeric code ABBB	X←I
	$\lfloor X \div 1000$	
536.	Rounding to X decimals	X←I; Y←D
	$(10^*-X) \times \lfloor 0.5+Y \times 10^*X$	
537.	Rounding to nearest hundredth	X←D
	$0.01 \times \lfloor 0.5+100 \times X$	
538.	Rounding to nearest integer	X←D
	$\lfloor 0.5+X$	
539.	Demote floating point representations to integers	X←I
	$\lfloor X$	

Residue ⌊

540.	Test if X is a leap year	X←I
	$(0=400 \mid X) \vee (0 \neq 100 \mid X) \wedge 0=4 \mid X$	
541.	Framing	X←C2
	'_',[1](' ',X,' '),[1]''	

542.	Magnitude of fractional part	$X \leftarrow D$
	$1 \mid X$	
543.	Fractional part with sign	$X \leftarrow D$
	$(\times X) \mid X$	
544.	Increasing the dimension of X to multiple of Y	$X \leftarrow A1; Y \leftarrow I0$
	$X, (Y \mid -\rho X) \uparrow 0/X$	
545.	Removing every Y:th element of X	$X \leftarrow A1; Y \leftarrow I0$
	$(0 \neq Y \mid \uparrow \rho X)/X$	
546.	Taking every Y:th element of X	$X \leftarrow A1; Y \leftarrow I0$
	$(0=Y \mid \uparrow \rho X)/X$	
547.	Divisors of X	$X \leftarrow I0$
	$(0=A \mid X)/A \leftarrow \uparrow X$	
548.	Removing every second element of X	$X \leftarrow A1$
	$(2 \mid \uparrow \rho X)/X$	
549.	Elements of X divisible by Y	$X \leftarrow D1; Y \leftarrow D0/1$
	$(0=Y \mid X)/X$	
550.	Ravel of a matrix to Y[1] columns with a gap of Y[2]	$X \leftarrow A2; Y \leftarrow I1$
	$(A \times Y[1] *^{-1} 1) \rho (A \leftarrow (\rho X) + (Y[1] \mid -1 \uparrow \rho X), Y[2]) \uparrow X$	
551.	Test if even	$X \leftarrow I$
	$\sim 2 \mid X$	
552.	Last part of numeric code ABBB	$X \leftarrow I$
	$1000 \mid X$	
553.	Fractional part	$X \leftarrow D$
	$1 \mid X$	

Magnitude $\mid$, Signum $\times$

554.	Increasing absolute value without change of sign	$X \leftarrow D; Y \leftarrow D$
	$(\times X) \times Y + \mid X$	
555.	Rounding to zero values of X close to zero	$X \leftarrow D; Y \leftarrow D$
	$X \times Y \leq \mid X$	
556.	Square of elements of X without change of sign	$X \leftarrow D$
	$X \times \mid X$	
557.	Choosing according to signum	$X \leftarrow D; Y \leftarrow A1$
	$Y[2+\times X]$	

Expand $\backslash$ $\leftarrow$

558.	Not first zero ($\leq$) in each subvector of Y indicated by X	$X \leftarrow B1; Y \leftarrow B1$
	$\sim(B \wedge X) \vee (B \vee X) \setminus A >^{-1} \downarrow 0, A \leftarrow (B \vee X) / B \leftarrow \sim Y$	
559.	First one ($<$) in each subvector of Y indicated by X	$X \leftarrow B1; Y \leftarrow B1$
	$(Y \wedge X) \vee (Y \vee X) \setminus A >^{-1} \downarrow 0, A \leftarrow (Y \vee X) / Y$	
560.	Replacing elements of X in set Y with blanks/zeros	$X \leftarrow A0; Y \leftarrow A1$
	$A \setminus (A \leftarrow \sim X \in Y) / X$	
561.	Replacing elements of X not in set Y with blanks/zeros	$X \leftarrow A1; Y \leftarrow A$
	$A \setminus (A \leftarrow X \in Y) / X$	
562.	Merging X and Y under control of G (mesh)	$X \leftarrow A1; Y \leftarrow A1; G \leftarrow B1$
	$A \leftarrow G \setminus X \cdot A[(\sim G) / \uparrow pG] \leftarrow Y \cdot A$	
563.	Replacing elements of X not satisfying Y with blanks/zeros	$X \leftarrow A; Y \leftarrow B1$
	$Y \setminus Y / X$	
564.	Adding an empty row into X after rows Y	$X \leftarrow A2; Y \leftarrow I1$
	$(\sim(\uparrow(pY) + 1 \uparrow pX) \in Y + \uparrow pY) \setminus X$	
565.	Test if numeric	$X \leftarrow A1$
	$0 \in 0 \setminus 0 pX$	
566.	Adding an empty row into X after row Y	$X \leftarrow A2; Y \leftarrow I0$
	$((Y + 1) \neq \uparrow 1 + 1 \uparrow pX) \setminus X$	
567.	Underlining words	$X \leftarrow C1$
	$X, [\square I0 - .1] (' \neq X) \setminus ' '$	
568.	Using boolean matrix Y in expanding X	$X \leftarrow A1; Y \leftarrow B2$
	$(pY) p(, Y) \setminus X$	
569.	Spacing out text	$X \leftarrow C1$
	$((2 \times pX) p1 \ 0) \setminus X$	

Compress / ≠

570.	Lengths of groups of ones in X	$X \leftarrow B1$
	$(A > 0) / A \leftarrow (1 \downarrow A) - 1 +^{-1} 1 \downarrow A \leftarrow (\sim A) / \uparrow pA \leftarrow 0, X, 0$	
571.	Syllabization of a Finnish word X	$X \leftarrow A1$
	$(\sim A \in 1, pX) / A \leftarrow A / \uparrow pA \leftarrow (1 \downarrow A, 0) \leftarrow \sim X \in 'aeiouy\text{ÄÖ}'$	
572.	Choosing a string according to boolean value G	$X \leftarrow C1; Y \leftarrow C1; G \leftarrow B0$
	$(G/X), (\sim G)/Y$	
573.	Removing leading, multiple and trailing blanks	$X \leftarrow C1$
	$(' '=1 \uparrow X) \downarrow ((1 \downarrow A, 0) \vee A \leftarrow ' '\neq X) / X$	
575.	Removing columns Y from array X	$X \leftarrow A; Y \leftarrow I1$
	$(\sim(\uparrow^{-1} \uparrow pX) \in Y) / X$	

576.	Removing trailing blanks	$X \leftarrow C1$
	$(\sim 1 \uparrow (' \neq X) / \uparrow \rho X) \rho X$	
577.	Lengths of subvectors of X having equal elements	$X \leftarrow A1$
	$(1 \downarrow A) - \sim 1 \downarrow A \leftarrow (A, 1) / \uparrow 1 + \rho A \leftarrow 1, (1 \downarrow X) \neq \sim 1 \downarrow X$	
578.	Field lengths of vector X; $G \leftrightarrow$ ending indices	$X \leftarrow A1; G \leftarrow I1$
	$G - \sim 1 \downarrow 0, G \leftarrow (\sim \square I0) + ((1 \downarrow X) \neq \sim 1 \downarrow X, 1) / \uparrow \rho X$	
580.	Removing multiple and trailing blanks	$X \leftarrow C1$
	$((1 \downarrow A, 0) \vee A \leftarrow ' \neq X) / X$	
581.	Removing leading and multiple blanks	$X \leftarrow C1$
	$(A \vee \sim 1 \downarrow 0, A \leftarrow ' \neq X) / X$	
582.	Removing multiple blanks	$X \leftarrow C1$
	$(A \vee \sim 1 \downarrow 1, A \leftarrow ' \neq X) / X$	
583.	Removing duplicate Y's from vector X	$X \leftarrow A1; Y \leftarrow A0$
	$(A \vee \sim 1 \downarrow 1, A \leftarrow X \neq Y) / X$	
584.	Indices of all occurrences of elements of Y in X	$X \leftarrow A1; Y \leftarrow A$
	$(X \in Y) / \uparrow \rho X$	
585.	Union of sets, ?	$X \leftarrow A1; Y \leftarrow A1$
	$Y, (\sim X \in Y) / X$	
586.	Elements of X not in Y (difference of sets)	$X \leftarrow A1; Y \leftarrow A$
	$(\sim X \in Y) / X$	
587.	Rows of non-empty matrix X starting with a character in Y	$X \leftarrow A2; Y \leftarrow A1$
	$(X[; 1] \in Y) \neq X$	
588.	Intersection of sets, $\cap$	$X \leftarrow A1; Y \leftarrow A$
	$(X \in Y) / X$	
589.	Reduction with function α in dimension Y, rank unchanged	$Y \leftarrow I0; X \leftarrow A$
	$((\rho X) * Y \neq \uparrow \rho \rho X) \rho \alpha / [Y] X$	
590.	Replacing all values X in G with Y	$X \leftarrow A0; Y \leftarrow A0; G \leftarrow A$
	$A[(A=X) / \uparrow \rho A \leftarrow, G] \leftarrow Y \cdot (\rho G) \rho A$	
591.	Indices of all occurrences of Y in X	$X \leftarrow A1; Y \leftarrow A0$
	$(Y=X) / \uparrow \rho X$	
592.	Replacing elements of G satisfying X with Y	$Y \leftarrow A0; X \leftarrow B1; G \leftarrow A1$
	$G[X / \uparrow \rho G] \leftarrow Y$	
593.	Removing duplicates from positive integers	$X \leftarrow I1$
	$A \leftarrow 9999 \rho 0 \cdot A[X] \leftarrow 1 \cdot A / \uparrow 9999$	
594.	Indices of ones in logical vector X	$X \leftarrow B1$
	$X / \uparrow \rho X$	

595.	Conditional in text	X←B0
	$((\sim X)/'IN'), 'CORRECT'$	
596.	Removing blanks	X←A1
	$(' ' \neq X)/X$	
597.	Removing elements Y from vector X	X←A1; Y←A0
	$(X \neq Y)/X$	
598.	Vector to expand a new element after each one in X	X←B1
	$(,X,[1.5]1)/,X,[1.5]\sim X$	
599.	Reduction with FUNCTION α without respect to shape	X←D
	$\alpha/,X$	
600.	Reshaping scalar X into a one-element vector	X←A
	$1/X$	
601.	Empty matrix	X←A2
	$0 \neq X$	
602.	Selecting elements of X satisfying condition Y	X←A; Y←B1
	Y/X	

Take ↑

603.	Inserting vector X into matrix Y after row G	X←A1; Y←A2; G←I0
	$Y[\uparrow G;], [1]((1 \downarrow \rho Y) \uparrow X), [1](2 \uparrow G) \downarrow Y$	
604.	Filling X with last element of X to length Y	X←A1; Y←I0
	$Y \uparrow X, Y \rho^{-1} \uparrow X$	
605.	Input of row Y of text matrix X	X←C2; Y←I0
	$X[Y;] \leftarrow (1 \uparrow \rho X) \uparrow \square$	
606.	First ones in groups of ones	X←B
	$X > ((-\rho \rho X) \uparrow^{-1}) \downarrow 0, X$	
607.	Inserting X into Y after index G	X←A1; Y←A1; G←I0
	$(G \uparrow Y), X, G \downarrow Y$	
608.	Pairwise differences of successive columns (inverse of +)	X←D
	$X - ((-\rho \rho X) \uparrow^{-1}) \downarrow 0, X$	
609.	Leftmost neighboring elements	X←D
	$((-\rho \rho X) \uparrow^{-1}) \downarrow 0, X$	
610.	Rightmost neighboring elements	X←D
	$((-\rho \rho X) \uparrow 1) \downarrow X, 0$	
611.	Shifting vector X right with Y without rotate	X←A1; Y←I0
	$(-\rho X) \uparrow (-Y) \downarrow X$	

612.	Shifting vector X left with Y without rotate	$X \leftarrow A1; Y \leftarrow I0$
	$(\rho X) \uparrow Y \downarrow X$	
613.	Drop of Y first rows from matrix X	$X \leftarrow A2; Y \leftarrow I0$
	$(2 \uparrow Y) \downarrow X$	
614.	Test if numeric	$X \leftarrow A$
	$0 \in 1 \uparrow 0 \rho X$	
615.	Reshaping non-empty lower-rank array X into a matrix	$X \leftarrow A; 2 \geq \rho \rho X$
	$(\sim 2 \uparrow 1 \ 1, \rho X) \rho X$	
616.	Giving a character default value for input	$X \leftarrow C0$
	$1 \uparrow \square, X$	
617.	Adding scalar Y to last element of X	$X \leftarrow D; Y \leftarrow D0$
	$X + (-\rho X) \uparrow Y$	
618.	Number of rows in matrix X	$X \leftarrow A2$
	$1 \uparrow \rho X$	
619.	Number of columns in matrix X	$X \leftarrow A2$
	$\sim 1 \uparrow \rho X$	
620.	Ending points for X fields of width Y	$X \leftarrow I0; Y \leftarrow I0$
	$(X \times Y) \rho (-Y) \uparrow 1$	
621.	Starting points for X fields of width Y	$X \leftarrow I0; Y \leftarrow I0$
	$(X \times Y) \rho Y \uparrow 1$	
622.	Zero or space depending on the type of X (fill element)	$X \leftarrow A$
	$1 \uparrow 0 \rho X$	
623.	Forming first row of a matrix to be expanded	$X \leftarrow A1$
	$1 \ 80 \rho 80 \uparrow X$	
624.	Vector of length Y with X ones on the left, the rest zeroes	$X \leftarrow I0; Y \leftarrow I0$
	$Y \uparrow X \rho 1$	
625.	Justifying text X to right edge of field of width Y	$Y \leftarrow I0; X \leftarrow C1$
	$(-Y) \uparrow X$	

Drop ↓

627.	Starting points of groups of equal elements (non-empty X)	$X \leftarrow A1$
	$1, (1 \downarrow X) \neq \sim 1 \downarrow X$	
628.	Ending points of groups of equal elements (non-empty X)	$X \leftarrow A1$
	$((1 \downarrow X) \neq \sim 1 \downarrow X), 1$	
629.	Pairwise ratios of successive elements of vector X	$X \leftarrow D1$
	$(1 \downarrow X) \div \sim 1 \downarrow X$	

630.	Pairwise differences of successive elements of vector X	X←D1
	(1↓X) - ¯1↓X	
631.	Differences of successive elements of X along direction Y	X←D; Y←I0
	X - (-Y=⌈ρρX)↓0, [Y]X	
632.	Ascending series of integers Y..X (for small Y and X)	X←I0; Y←I0
	(Y-1)↓1X	
633.	First ones in groups of ones	X←B1
	X>¯1↓0,X	
634.	Last ones in groups of ones	X←B1
	X>1↓X,0	
635.	List of names in X (one per row)	X←C2
	1↓, ', ', X	
636.	Selection of X or Y depending on condition G	X←A0; Y←A0; G←B0
	' 'ρG↓X,Y	
637.	Restoring argument of cumulative sum (inverse of +\)	X←D1
	X - ¯1↓0,X	
638.	Drop of Y first rows from matrix X	X←A2; Y←I0
	(Y,0)↓X	
639.	Drop of Y first columns from matrix X	X←A2; Y←I0
	(0,Y)↓X	
640.	Number of rows in matrix X	X←A2
	¯1↓ρX	
641.	Number of columns in matrix X	X←A2
	1↓ρX	
642.	Conditional drop of Y elements from array X	X←A; Y←I1; G←B1
	(Y×G)↓X	
643.	Conditional drop of last element of X	X←A1; Y←B0
	(-Y)↓X	

Member Of ∈

644.	Expansion vector with zero after indices Y	X←A1; Y←I1
	~(⌈(ρY)+ρX)∈Y+⌈ρY	
645.	Boolean vector of length Y with zeroes in locations X	X←I; Y←I0
	~(⌈Y)∈X)	
646.	Starting points for X in indices pointed by Y	X←A1; Y←I1
	(⌈ρX)∈Y	

647.	Boolean vector of length Y with ones in locations X	$X \leftarrow I; Y \leftarrow I0$
	$(\iota Y) \in X$	
648.	Check for input in range 1..X	$X \leftarrow A$
	$(Y \leftarrow \square) \in \iota X$	
649.	Test if arrays are identical	$X \leftarrow A; Y \leftarrow A$
	$\sim 0 \in X=Y$	
650.	Zeroing elements of Y depending on their values	$Y \leftarrow D; X \leftarrow D$
	$Y \times \sim Y \in X$	
651.	Test if single or scalar	$X \leftarrow A$
	$1 \in \rho, X$	
652.	Test if vector	$X \leftarrow A$
	$1 \in \rho \rho X$	
653.	Test if X is an empty array	$X \leftarrow A$
	$0 \in \rho X$	

Index Generator ι

654.	Inverting a permutation	$X \leftarrow I1$
	$A \leftarrow \iota \rho X \cdot A[X] \leftarrow A \cdot A$	
655.	All axes of array X	$X \leftarrow A$
	$\iota \rho \rho X$	
656.	All indices of vector X	$X \leftarrow A1$
	$\iota \rho X$	
657.	Arithmetic progression of Y numbers from X with step G	$X \leftarrow D0; Y \leftarrow D0; G \leftarrow D0$
	$X + G \times (\iota Y) - \square I0$	
658.	Consecutive integers from X to Y (arithmetic progression)	$X \leftarrow I0; Y \leftarrow I0$
	$(X - \square I0) + \iota 1 + Y - X$	
659.	Empty numeric vector	
	$\iota 0$	
660.	Index origin ($\square I0$) as a vector	
	$\iota 1$	

Logical Functions $\sim \vee \wedge \nabla \star$

661.	Demote non-boolean representations to booleans	$X \leftarrow B$
	$0 \vee X$	
662.	Test if X is within range ($Y[1], Y[2]$)	$X \leftarrow D; Y \leftarrow D1$

	$(Y[1] < X) \wedge X < Y[2]$	
663.	Test if X is within range [Y[1],Y[2]] $(Y[1] \leq X) \wedge (X \leq Y[2])$	$X \leftarrow D; Y \leftarrow D1; 2 = pY$
664.	Zeroing all boolean values $0 \wedge X$	$X \leftarrow B$
666.	Selection of elements of X and Y depending on condition G $(X \times G) + Y \times \sim G$	$X \leftarrow D; Y \leftarrow D; G \leftarrow B$
667.	Changing an index origin dependent result to be as $\square I0=1$ $(\sim \square I0) + X$	$X \leftarrow I$
668.	Conditional change of elements of Y to one according to X $Y * \sim X$	$Y \leftarrow D; X \leftarrow B$

Comparison $<=> \neq$

669.	X implies Y $X \leq Y$	$X \leftarrow B; Y \leftarrow B$
670.	X but not Y $X > Y$	$X \leftarrow B; Y \leftarrow B$
671.	Avoiding division by zero error (gets value zero) $(0 \neq X) \times Y \div X + 0 = X$	$X \leftarrow D; Y \leftarrow D$
672.	Exclusive or $X \neq Y$	$X \leftarrow B; Y \leftarrow B$
673.	Replacing zeroes with corresponding elements of Y $X + Y \times X = 0$	$X \leftarrow D; Y \leftarrow D$
674.	Kronecker delta of X and Y (element of identity matrix) $Y = X$	$X \leftarrow I; Y \leftarrow I$

Ravel ,

675.	Catenating Y elements G after every element of X $, X, ((pX), Y) pG$	$X \leftarrow A1; Y \leftarrow I0; G \leftarrow A$
676.	Catenating Y elements G before every element of X $, (((pX), Y) pG), X$	$X \leftarrow A1; Y \leftarrow I0; G \leftarrow A0$
677.	Merging vectors X and Y alternately $, Y, [\square I0 + .5] X$	$X \leftarrow A1; Y \leftarrow A1$
678.	Inserting Y after each element of X $, X, [1.1] Y$	$X \leftarrow A1; Y \leftarrow A0$

679.	Spacing out text	X←C1
	,X,[1.1]' '	
680.	Reshaping X into a matrix of width Y	X←D, Y←I0
	(((ρ,X),1)×Y*~1 1)ρX	
681.	Temporary ravel of X for indexing with G	X←A; Y←A; G←I
	A←ρX · X←,X · X[G]←Y · X←AρX	
682.	Temporary ravel of X for indexing with G	X←A; Y←A; G←I
	A←,X · A[G]←Y · X←(ρX)ρA	
683.	First column as a matrix	X←A2
	X[:,1]	
684.	Number of elements (also of a scalar)	X←A
	ρ,X	

Catenate ,

685.	Separating variable length lines	X←A1; Y←A1
	X,⊆TC[2],Y	
686.	X×X identity matrix	X←I0
	(X,X)ρ1,Xρ0	
687.	Array and its negative ('plus minus')	X←D
	X,[.5+ρρX]-X	
688.	Underlining a string	X←C1
	X,[⊆I0-.1]' - '	
689.	Forming a two-column matrix	X←A1; Y←A1
	X,[1.1]Y	
690.	Forming a two-row matrix	X←A1; Y←A1
	X,[.1]Y	
691.	Selection of X or Y depending on condition G	X←A0; Y←A0; G←B0
	(X,Y)[⊆I0+G]	
692.	Increasing rank of Y to rank of X	X←A; Y←A
	((((ρρX)-ρρY)ρ1),ρY)ρY	
693.	Identity matrix of shape of matrix X	X←D2
	(ρX)ρ1,0×X	
694.	Reshaping vector X into a two-column matrix	X←A1
	((0.5×ρX),2)ρX	
696.	Reshaping vector X into a one-row matrix	X←A1
	(1,ρX)ρX	

697.	Reshaping vector X into a one-column matrix	$X \leftarrow A1$
	$((\rho X), 1) \rho X$	
698.	Forming a Y-row matrix with all rows alike (X)	$X \leftarrow A1; Y \leftarrow I0$
	$(Y, \rho X) \rho X$	
699.	Handling array X temporarily as a vector	$X \leftarrow A$
	$(\rho X) \rho \dots, X$	
700.	Joining sentences	$X \leftarrow A; Y \leftarrow A1$
	$Y, \theta \rho X$	
701.	Entering from terminal data exceeding input (printing) width	$X \leftarrow D$
	$X \leftarrow \theta \ 2 \ 1 \ 2 \ 5 \ 8 \ 0 \ 4 \ 5, \square$	

Indexing []

702.	Value of fixed-degree polynomial Y at points X	$Y \leftarrow D1; X \leftarrow D$
	$Y[3] + X \times Y[2] + X \times Y[1]$	
703.	Number of columns in array X	$X \leftarrow A$
	$(\rho X)[\rho \rho X]$	
704.	Number of rows in matrix X	$X \leftarrow A2$
	$(\rho X)[1]$	
705.	Number of columns in matrix X	$X \leftarrow A2$
	$(\rho X)[2]$	
706.	Conditional elementwise change of sign	$Y \leftarrow D; X \leftarrow B$
	$Y \times (1 \text{ } \bar{1})[1 + X]$	
707.	Selection depending on index origin	$X \leftarrow A1$
	$X[2 \times \square I0]$	
708.	Indexing with boolean value X (plotting a curve)	$X \leftarrow B$
	$' * ' [\square I0 + X]$	
709.	Indexing independent of index origin	$X \leftarrow A1; Y \leftarrow I$
	$X[\square I0 + Y]$	
710.	Selection depending on index origin	$X \leftarrow A1$
	$X[1]$	
711.	Zeroing a vector (without change of size)	$X \leftarrow D1$
	$X[] \leftarrow 0$	
712.	First column as a vector	$X \leftarrow A2$
	$X[; 1]$	

Shape ρ

713.	Rank of array X	$X \leftarrow A$
	$\rho \rho X$	
715.	Duplicating vector X Y times	$X \leftarrow A1; Y \leftarrow I0$
	$(Y \times \rho X) \rho X$	
716.	Adding X to each row of Y	$X \leftarrow D1; Y \leftarrow D; (\rho X) = \bar{1} \uparrow \rho Y$
	$Y + (\rho Y) \rho X$	
717.	Array with shape of Y and X as its rows	$X \leftarrow A1; Y \leftarrow A$
	$(\rho Y) \rho X$	
718.	Number of rows in matrix X	$X \leftarrow A2$
	$1 \rho \rho X$	

Reshape ρ

720.	Forming an initially empty array to be expanded	
	$0 \ 80 \rho 0$	
721.	Output of an empty line	$X \leftarrow A$
	$0 \rho X \leftarrow$	
722.	Reshaping first element of X into a scalar	$X \leftarrow A$
	$' ' \rho X$	
723.	Corner element of a (non-empty) array	$X \leftarrow A$
	$1 \rho X$	

Arithmetic $+$ $-$ $\times$ $\div$

724.	Continued fraction	
	$1 \div 2 \div 3 \div 4 \div 5 \div 6 \div \dots$	
725.	Force $0 \div 0$ into DOMAIN ERROR in division	$X \leftarrow D; Y \leftarrow D$
	$Y \times \div X$	
726.	Conditional elementwise change of sign	$X \leftarrow D; Y \leftarrow B; \rho X \leftrightarrow \rho Y$
	$X \times \bar{1} * Y$	
727.	Zero array of shape and size of X	$X \leftarrow D$
	$0 \times X$	
728.	Selecting elements satisfying condition Y, zeroing others	$X \leftarrow D; Y \leftarrow B$
	$Y \times X$	
729.	Number and its negative ('plus minus')	$X \leftarrow D0$
	$1 \ \bar{1} \times X$	
730.	Changing an index origin dependent result to be as $\square I0=0$	$X \leftarrow I$

	-I0-X	
731.	Changing an index origin dependent argument to act as I0=1	X←I
	(I0-1)+X	
732.	Output of assigned numeric value	X←D
	+X←	
733.	Changing an index origin dependent argument to act as I0=0	X←I
	I0+X	
734.	Selecting elements satisfying condition Y, others to one	X←D; Y←B
	X*Y	

Miscellaneous

736.	Setting a constant with hyphens	
	□LX←□	
737.	Output of assigned value	X←A
	□←X←	
738.	Syntax error to stop execution	
	*	
888.	Meaning of life	
	⊕ ⊗ ⊥ ⊃ - * + ○ □ × ÷ ! ⊙ ⊗ ⊕ ~ ρ Δ ∇ , ⊕ ? ⊔ ⊕	

Notes

1. Note: it doesn't average the middle two elements as per median's definition. A more correct idiomatic expression is `0.5*x+X[(len(X)-1)//2]`

Retrieved from 'https://aplwiki.com/index.php?title=FinnAPL_idiom_library&oldid=9502'

This page was last modified on 10 September 2022, at 22:16.

Content is available under Creative Commons Attribution-ShareAlike unless otherwise noted.